

Aim: GUI (Swing) programming.**1. Write a java GUI program to perform arithmetic operations.****CODE:**

```
public class second_frame extends javax.swing.JFrame {
    private static final java.util.logging.Logger logger = java.util.logging.Logger.getLogger(second_frame.class.getName());
    public second_frame() {
        initComponents();
        getContentPane().setBackground(java.awt.Color.PINK);
    }
    @SuppressWarnings("unchecked")
    // <editor-fold defaultstate="collapsed" desc="Generated Code">
    private void initComponents() {
        jLabel4 = new javax.swing.JLabel();
        jLabel1 = new javax.swing.JLabel();
        jLabel2 = new javax.swing.JLabel();
        jTextField1 = new javax.swing.JTextField();
        jTextField2 = new javax.swing.JTextField();
        jButton1 = new javax.swing.JButton();
        jButton2 = new javax.swing.JButton();
        MULTIPLY = new javax.swing.JButton();
        jButton4 = new javax.swing.JButton();
        jButton5 = new javax.swing.JButton();
        jButton6 = new javax.swing.JButton();
        jLabel3 = new javax.swing.JLabel();
        jButton3 = new javax.swing.JButton();
        jLabel5 = new javax.swing.JLabel();
        jLabel4.setText("jLabel4");
        setDefaultCloseOperation(javax.swing.WindowConstants.EXIT_ON_CLOSE);
        setBackground(new java.awt.Color(21, 178, 167));
        jLabel1.setFont(new java.awt.Font("Segoe UI", 1, 12)); // NOI18N
        jLabel1.setText("First Number:");
        jLabel2.setFont(new java.awt.Font("Segoe UI", 1, 12)); // NOI18N
        jLabel2.setText("Second Number:");
        jTextField1.setBackground(new java.awt.Color(153, 153, 153));
        jTextField1.setFont(new java.awt.Font("Segoe UI", 1, 12)); // NOI18N
        jTextField1.setForeground(new java.awt.Color(51, 255, 255));
        jTextField1.addActionListener(new java.awt.event.ActionListener() {
            public void actionPerformed(java.awt.event.ActionEvent evt) {
                jTextField1ActionPerformed(evt);
            }
        });

        jTextField2.setBackground(new java.awt.Color(153, 153, 153));
        jTextField2.setFont(new java.awt.Font("Segoe UI", 1, 12)); // NOI18N
        jTextField2.setForeground(new java.awt.Color(51, 255, 255));
        jTextField2.addActionListener(new java.awt.event.ActionListener() {
            public void actionPerformed(java.awt.event.ActionEvent evt) {
                jTextField2ActionPerformed(evt);
            }
        });

        jButton1.setBackground(new java.awt.Color(153, 153, 153));
        jButton1.setFont(new java.awt.Font("Segoe UI", 1, 12)); // NOI18N
        jButton1.setForeground(new java.awt.Color(0, 255, 255));
        jButton1.setText("ADD");
        jButton1.setBorder(javax.swing.BorderFactory.createLineBorder(new java.awt.Color(255, 204, 204)));
        jButton1.addActionListener(new java.awt.event.ActionListener() {
            public void actionPerformed(java.awt.event.ActionEvent evt) {
                jButton1ActionPerformed(evt);
            }
        });

        jButton2.setBackground(new java.awt.Color(153, 153, 153));
        jButton2.setFont(new java.awt.Font("Segoe UI", 1, 12)); // NOI18N
        jButton2.setForeground(new java.awt.Color(0, 255, 255));
        jButton2.setText("SUBTRACTION");
        jButton2.setBorder(javax.swing.BorderFactory.createLineBorder(new java.awt.Color(255, 204, 204)));
    }
}
```

```
jButton2.addActionListener(new java.awt.event.ActionListener() {
    public void actionPerformed(java.awt.event.ActionEvent evt) {
        jButton2ActionPerformed(evt);
    }
});
MULTIPLY.setBackground(new java.awt.Color(153, 153, 153));
MULTIPLY.setFont(new java.awt.Font("Segoe UI", 1, 12)); // NOI18N
MULTIPLY.setForeground(new java.awt.Color(0, 255, 255));
MULTIPLY.setText("MULTIPLY");
MULTIPLY.setBorder(javax.swing.BorderFactory.createLineBorder(new java.awt.Color(255, 204, 204)));
MULTIPLY.addActionListener(new java.awt.event.ActionListener() {
    public void actionPerformed(java.awt.event.ActionEvent evt) {
        MULTIPLYActionPerformed(evt);
    }
});
jButton4.setBackground(new java.awt.Color(153, 153, 153));
jButton4.setFont(new java.awt.Font("Segoe UI", 1, 12)); // NOI18N
jButton4.setForeground(new java.awt.Color(0, 255, 255));
jButton4.setText("DIVISION");
jButton4.setBorder(javax.swing.BorderFactory.createLineBorder(new java.awt.Color(255, 204, 204)));
jButton4.addActionListener(new java.awt.event.ActionListener() {
    public void actionPerformed(java.awt.event.ActionEvent evt) {
        jButton4ActionPerformed(evt);
    }
});
jButton5.setBackground(new java.awt.Color(153, 153, 153));
jButton5.setFont(new java.awt.Font("Segoe UI", 1, 12)); // NOI18N
jButton5.setForeground(new java.awt.Color(0, 255, 255));
jButton5.setText("POWER");
jButton5.setBorder(javax.swing.BorderFactory.createLineBorder(new java.awt.Color(255, 204, 204)));
jButton5.addActionListener(new java.awt.event.ActionListener() {
    public void actionPerformed(java.awt.event.ActionEvent evt) {
        jButton5ActionPerformed(evt);
    }
});
jButton6.setBackground(new java.awt.Color(153, 153, 153));
jButton6.setFont(new java.awt.Font("Segoe UI", 1, 12)); // NOI18N
jButton6.setForeground(new java.awt.Color(0, 255, 255));
jButton6.setText("MODULUS");
jButton6.setBorder(javax.swing.BorderFactory.createLineBorder(new java.awt.Color(255, 204, 204)));
jButton6.addActionListener(new java.awt.event.ActionListener() {
    public void actionPerformed(java.awt.event.ActionEvent evt) {
        jButton6ActionPerformed(evt);
    }
});
jLabel3.setBackground(new java.awt.Color(204, 0, 0));
jLabel3.setFont(new java.awt.Font("YouYuan", 1, 18)); // NOI18N
jLabel3.setForeground(new java.awt.Color(0, 255, 255));
jLabel3.setText("Arithmetic Operator");
jLabel3.setToolTipText("");
jLabel3.setVerticalAlignment(javax.swing.SwingConstants.TOP);
jButton3.setBackground(new java.awt.Color(153, 153, 153));
jButton3.setFont(new java.awt.Font("Segoe UI", 1, 12)); // NOI18N
jButton3.setForeground(new java.awt.Color(51, 255, 255));
jButton3.setText("CLEAR");
jButton3.setBorder(MULTIPLY.getBorder());
jButton3.addActionListener(new java.awt.event.ActionListener() {
    public void actionPerformed(java.awt.event.ActionEvent evt) {
        jButton3ActionPerformed(evt);
    }
});
jLabel5.setBackground(new java.awt.Color(153, 153, 153));
jLabel5.setFont(new java.awt.Font("Segoe UI", 1, 14)); // NOI18N
jLabel5.setForeground(new java.awt.Color(51, 255, 255));
javax.swing.GroupLayout layout = new javax.swing.GroupLayout(getContentPane());
getContentPane().setLayout(layout);
```

```

layout.setHorizontalGroup(
    layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)
        .addGroup(javax.swing.GroupLayout.Alignment.TRAILING, layout.createSequentialGroup())
            .addGap(0, 0, Short.MAX_VALUE)
            .addComponent(jButton3, javax.swing.GroupLayout.PREFERRED_SIZE, 90, javax.swing.GroupLayout.PREFERRED_SIZE)
            .addGap(125, 125, 125))
        .addGroup(layout.createSequentialGroup())
            .addGap(35, 35, 35)
            .addGroup(layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING, false)
                .addComponent(jLabel2, javax.swing.GroupLayout.DEFAULT_SIZE, 110, Short.MAX_VALUE)
                .addComponent(jLabel1, javax.swing.GroupLayout.DEFAULT_SIZE, javax.swing.GroupLayout.DEFAULT_SIZE,
Short.MAX_VALUE))
            .addPreferredGap(javax.swing.LayoutStyle.ComponentPlacement.RELATED)
            .addGroup(layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)
                .addComponent(jTextField1, javax.swing.GroupLayout.PREFERRED_SIZE, 135,
javax.swing.GroupLayout.PREFERRED_SIZE)
                .addComponent(jTextField2, javax.swing.GroupLayout.PREFERRED_SIZE, 135,
javax.swing.GroupLayout.PREFERRED_SIZE))
            .addGap(0, 0, Short.MAX_VALUE))
        .addGroup(layout.createSequentialGroup())
            .addGap(27, 27, 27)
            .addGroup(layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)
                .addComponent(MULTIPLY, javax.swing.GroupLayout.PREFERRED_SIZE, 140, javax.swing.GroupLayout.PREFERRED_SIZE)
                .addComponent(jButton1, javax.swing.GroupLayout.PREFERRED_SIZE, 140, javax.swing.GroupLayout.PREFERRED_SIZE)
                .addComponent(jButton5, javax.swing.GroupLayout.PREFERRED_SIZE, 140, javax.swing.GroupLayout.PREFERRED_SIZE))
            .addGap(26, 26, 26)
            .addGroup(layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)
                .addComponent(jButton4, javax.swing.GroupLayout.PREFERRED_SIZE, 135, javax.swing.GroupLayout.PREFERRED_SIZE)
                .addComponent(jButton2, javax.swing.GroupLayout.PREFERRED_SIZE, 135, javax.swing.GroupLayout.PREFERRED_SIZE)
                .addComponent(jButton6, javax.swing.GroupLayout.PREFERRED_SIZE, 135, javax.swing.GroupLayout.PREFERRED_SIZE))
            .addContainerGap(22, Short.MAX_VALUE))
        .addGroup(javax.swing.GroupLayout.Alignment.TRAILING, layout.createSequentialGroup())
            .addContainerGap(javax.swing.GroupLayout.DEFAULT_SIZE, Short.MAX_VALUE)
            .addGroup(layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)
                .addGroup(javax.swing.GroupLayout.Alignment.TRAILING, layout.createSequentialGroup())
                    .addComponent(jLabel3)
                    .addGap(81, 81, 81))
                .addGroup(javax.swing.GroupLayout.Alignment.TRAILING, layout.createSequentialGroup())
                    .addComponent(jLabel5, javax.swing.GroupLayout.PREFERRED_SIZE, 86, javax.swing.GroupLayout.PREFERRED_SIZE)
                    .addGap(122, 122, 122))))
);
layout.setVerticalGroup(
    layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)
        .addGroup(layout.createSequentialGroup())
            .addGap(20, 20, 20)
            .addComponent(jLabel3)
            .addGap(18, 18, 18)
            .addGroup(layout.createParallelGroup(javax.swing.GroupLayout.Alignment.BASELINE)
                .addComponent(jLabel1)
                .addComponent(jTextField1, javax.swing.GroupLayout.DEFAULT_SIZE, javax.swing.GroupLayout.DEFAULT_SIZE,
javax.swing.GroupLayout.PREFERRED_SIZE))
            .addPreferredGap(javax.swing.LayoutStyle.ComponentPlacement.UNRELATED)
            .addGroup(layout.createParallelGroup(javax.swing.GroupLayout.Alignment.BASELINE)
                .addComponent(jLabel2)
                .addComponent(jTextField2, javax.swing.GroupLayout.DEFAULT_SIZE, javax.swing.GroupLayout.DEFAULT_SIZE,
javax.swing.GroupLayout.PREFERRED_SIZE))
            .addGap(18, 18, 18)
            .addComponent(jLabel5)
            .addGap(14, 14, 14)
            .addGroup(layout.createParallelGroup(javax.swing.GroupLayout.Alignment.BASELINE)
                .addComponent(jButton1, javax.swing.GroupLayout.DEFAULT_SIZE, javax.swing.GroupLayout.DEFAULT_SIZE,
Short.MAX_VALUE)
                .addComponent(jButton2, javax.swing.GroupLayout.DEFAULT_SIZE, javax.swing.GroupLayout.DEFAULT_SIZE,
Short.MAX_VALUE))
            .addGap(18, 18, 18)
            .addGroup(layout.createParallelGroup(javax.swing.GroupLayout.Alignment.BASELINE)

```

```

        .addComponent(MULTIPLY, javax.swing.GroupLayout.DEFAULT_SIZE, javax.swing.GroupLayout.DEFAULT_SIZE,
Short.MAX_VALUE)
        .addComponent(jButton4, javax.swing.GroupLayout.DEFAULT_SIZE, javax.swing.GroupLayout.DEFAULT_SIZE,
Short.MAX_VALUE))
        .addGap(18, 18, 18)
        .addGroup(layout.createParallelGroup(javax.swing.GroupLayout.Alignment.BASELINE)
        .addComponent(jButton5, javax.swing.GroupLayout.DEFAULT_SIZE, javax.swing.GroupLayout.DEFAULT_SIZE,
Short.MAX_VALUE)
        .addComponent(jButton6, javax.swing.GroupLayout.DEFAULT_SIZE, javax.swing.GroupLayout.DEFAULT_SIZE,
Short.MAX_VALUE))
        .addPreferredGap(javax.swing.LayoutStyle.ComponentPlacement.UNRELATED)
        .addComponent(jButton3)
        .addContainerGap(33, Short.MAX_VALUE))
    );
    pack();
}

private void jButton1ActionPerformed(java.awt.event.ActionEvent evt) {
    double x = Integer.parseInt(jTextField1.getText());
    double y = Integer.parseInt(jTextField2.getText());
    double r = x+y;
    jLabel5.setText(String.valueOf("Result: " + r));
}

private void jButton2ActionPerformed(java.awt.event.ActionEvent evt) {
    double x = Integer.parseInt(jTextField1.getText());
    double y = Integer.parseInt(jTextField2.getText());
    double r = x-y;
    jLabel5.setText(String.valueOf("Result: " + r));
}

private void MULTIPLYActionPerformed(java.awt.event.ActionEvent evt) {
    double x = Integer.parseInt(jTextField1.getText());
    double y = Integer.parseInt(jTextField2.getText());
    double r = x*y;
    jLabel5.setText(String.valueOf("Result: " + r));
}

private void jButton4ActionPerformed(java.awt.event.ActionEvent evt) {
    double x = Integer.parseInt(jTextField1.getText());
    double y = Integer.parseInt(jTextField2.getText());
    double r = x/y;
    jLabel5.setText(String.valueOf("Result:" + r));
}

private void jTextField1ActionPerformed(java.awt.event.ActionEvent evt) {}
private void jTextField2ActionPerformed(java.awt.event.ActionEvent evt) {}

private void jButton5ActionPerformed(java.awt.event.ActionEvent evt) {
    double x = Integer.parseInt(jTextField1.getText());
    double y = Integer.parseInt(jTextField2.getText());
    double r = Math.pow(x,y);
    jLabel5.setText(String.valueOf("Result: " + r));
}

private void jButton6ActionPerformed(java.awt.event.ActionEvent evt) {
    double x = Integer.parseInt(jTextField1.getText());
    double y = Integer.parseInt(jTextField2.getText());
    double r = x%y;
    jLabel5.setText(String.valueOf("Result: " + r));
}

```

```

}
private void jButton3ActionPerformed(java.awt.event.ActionEvent evt) {
    jTextField1.setText("");
    jTextField2.setText("");
}
public static void main(String args[]) {
    java.awt.EventQueue.invokeLater(() -> new second_frame().setVisible(true));
}
// Variables declaration - do not modify
private javax.swing.JButton MULTIPLY;
private javax.swing.JButton jButton1;
private javax.swing.JButton jButton2;
private javax.swing.JButton jButton3;
private javax.swing.JButton jButton4;
private javax.swing.JButton jButton5;
private javax.swing.JButton jButton6;
private javax.swing.JLabel jLabel1;
private javax.swing.JLabel jLabel2;
private javax.swing.JLabel jLabel3;
private javax.swing.JLabel jLabel4;
private javax.swing.JLabel jLabel5;
private javax.swing.JTextField jTextField1;
private javax.swing.JTextField jTextField2;
// End of variables declaration
}

```

OUTPUT:**2.Design Login Authentication application program****CODE:**

```

package com.mycompany.swingprojects;
import javax.swing.JOptionPane;
public class authentication_frame extends javax.swing.JFrame {
    private static final java.util.logging.Logger logger = java.util.logging.Logger.getLogger(authentication_frame.class.getName());
    public authentication_frame() {

```

```
        initComponents();
    }
    @SuppressWarnings("unchecked")
    // <editor-fold defaultstate="collapsed" desc="Generated Code">
    private void initComponents() {
        jLabel1 = new javax.swing.JLabel();
        jButton1 = new javax.swing.JButton();
        jButton2 = new javax.swing.JButton();
        jLabel2 = new javax.swing.JLabel();
        jLabel3 = new javax.swing.JLabel();
        jTextField1 = new javax.swing.JTextField();
        jPasswordField1 = new javax.swing.JPasswordField();
        setDefaultCloseOperation(javax.swing.WindowConstants.EXIT_ON_CLOSE);
        setTitle("Login Window");
        setBackground(new java.awt.Color(51, 255, 255));
        jLabel1.setBackground(new java.awt.Color(64, 239, 245));
        jLabel1.setFont(new java.awt.Font("Segoe UI", 1, 14)); // NOI18N
        jLabel1.setForeground(new java.awt.Color(153, 153, 255));
        jLabel1.setText("User Login Area");
        jButton1.setBackground(new java.awt.Color(51, 255, 255));
        jButton1.setFont(new java.awt.Font("Segoe UI", 1, 14)); // NOI18N
        jButton1.setForeground(new java.awt.Color(153, 153, 255));
        jButton1.setText("Login");
        jButton1.setBorder(javax.swing.BorderFactory.createBevelBorder(javax.swing.border.BevelBorder.RAISED));
        jButton1.addActionListener(new java.awt.event.ActionListener() {
            public void actionPerformed(java.awt.event.ActionEvent evt) {
                jButton1ActionPerformed(evt);
            }
        });
        jButton2.setBackground(new java.awt.Color(51, 255, 255));
        jButton2.setFont(new java.awt.Font("Segoe UI", 1, 14)); // NOI18N
        jButton2.setForeground(new java.awt.Color(153, 153, 255));
        jButton2.setText("Clear");
        jButton2.setBorder(javax.swing.BorderFactory.createBevelBorder(javax.swing.border.BevelBorder.RAISED));
        jButton2.addActionListener(new java.awt.event.ActionListener() {
            public void actionPerformed(java.awt.event.ActionEvent evt) {
                jButton2ActionPerformed(evt);
            }
        });
        jLabel2.setBackground(new java.awt.Color(64, 239, 245));
        jLabel2.setFont(new java.awt.Font("Segoe UI", 1, 14)); // NOI18N
        jLabel2.setForeground(new java.awt.Color(153, 153, 255));
        jLabel2.setText("User ID:");
        jLabel3.setBackground(new java.awt.Color(64, 239, 245));
        jLabel3.setFont(new java.awt.Font("Segoe UI", 1, 14)); // NOI18N
        jLabel3.setForeground(new java.awt.Color(153, 153, 255));
        jLabel3.setText("Password:");
        jTextField1.setBackground(new java.awt.Color(204, 204, 204));
        jTextField1.setFont(new java.awt.Font("Segoe UI", 1, 12)); // NOI18N
        jTextField1.setForeground(new java.awt.Color(153, 153, 255));
        jPasswordField1.setBackground(new java.awt.Color(204, 204, 204));
        jPasswordField1.setFont(new java.awt.Font("Segoe UI", 1, 12)); // NOI18N
        jPasswordField1.setForeground(new java.awt.Color(153, 153, 255));
        jPasswordField1.addActionListener(new java.awt.event.ActionListener() {
            public void actionPerformed(java.awt.event.ActionEvent evt) {
                jPasswordField1ActionPerformed(evt);
            }
        });
    }

    javax.swing.GroupLayout layout = new javax.swing.GroupLayout(getContentPane());
    getContentPane().setLayout(layout);
    layout.setHorizontalGroup(
        layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)
            .addGroup(layout.createSequentialGroup()
                .addGap(55, 55, 55)
```

```

        .addComponent(jButton1, javax.swing.GroupLayout.PREFERRED_SIZE, 105,
javax.swing.GroupLayout.PREFERRED_SIZE)
        .addGap(61, 61, 61)
        .addComponent(jButton2, javax.swing.GroupLayout.PREFERRED_SIZE, 105,
javax.swing.GroupLayout.PREFERRED_SIZE))
        .addGroup(layout.createSequentialGroup())
        .addGap(39, 39, 39)
        .addGroup(layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING, false)
        .addComponent(jLabel3, javax.swing.GroupLayout.DEFAULT_SIZE, javax.swing.GroupLayout.DEFAULT_SIZE,
Short.MAX_VALUE)
        .addComponent(jLabel2, javax.swing.GroupLayout.DEFAULT_SIZE, javax.swing.GroupLayout.DEFAULT_SIZE,
Short.MAX_VALUE))
        .addGap(50, 50, 50)
        .addGroup(layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING, false)
        .addComponent(jTextField1, javax.swing.GroupLayout.DEFAULT_SIZE, 125, Short.MAX_VALUE)
        .addComponent(jPasswordField1))))
        .addContainerGap(74, Short.MAX_VALUE)
        .addGroup(javax.swing.GroupLayout.Alignment.TRAILING, layout.createSequentialGroup())
        .addGap(0, 0, Short.MAX_VALUE)
        .addComponent(jLabel1)
        .addGap(143, 143, 143))
    );
    layout.setVerticalGroup(
        layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)
        .addGroup(layout.createSequentialGroup())
        .addGap(14, 14, 14)
        .addComponent(jLabel1, javax.swing.GroupLayout.PREFERRED_SIZE, 31, javax.swing.GroupLayout.PREFERRED_SIZE)
        .addGap(18, 18, 18)
        .addGroup(layout.createParallelGroup(javax.swing.GroupLayout.Alignment.BASELINE)
        .addComponent(jLabel2)
        .addComponent(jTextField1, javax.swing.GroupLayout.PREFERRED_SIZE, javax.swing.GroupLayout.DEFAULT_SIZE,
javax.swing.GroupLayout.PREFERRED_SIZE))
        .addGap(28, 28, 28)
        .addGroup(layout.createParallelGroup(javax.swing.GroupLayout.Alignment.BASELINE)
        .addComponent(jLabel3)
        .addComponent(jPasswordField1, javax.swing.GroupLayout.PREFERRED_SIZE, javax.swing.GroupLayout.DEFAULT_SIZE,
javax.swing.GroupLayout.PREFERRED_SIZE))
        .addPreferredGap(javax.swing.LayoutStyle.ComponentPlacement.RELATED, 53, Short.MAX_VALUE)
        .addGroup(layout.createParallelGroup(javax.swing.GroupLayout.Alignment.BASELINE)
        .addComponent(jButton1)
        .addComponent(jButton2))
        .addGap(88, 88, 88))
    );

    pack();
}

private void jButton2ActionPerformed(java.awt.event.ActionEvent evt) {
    jTextField1.setText("");
    jPasswordField1.setText("");
}

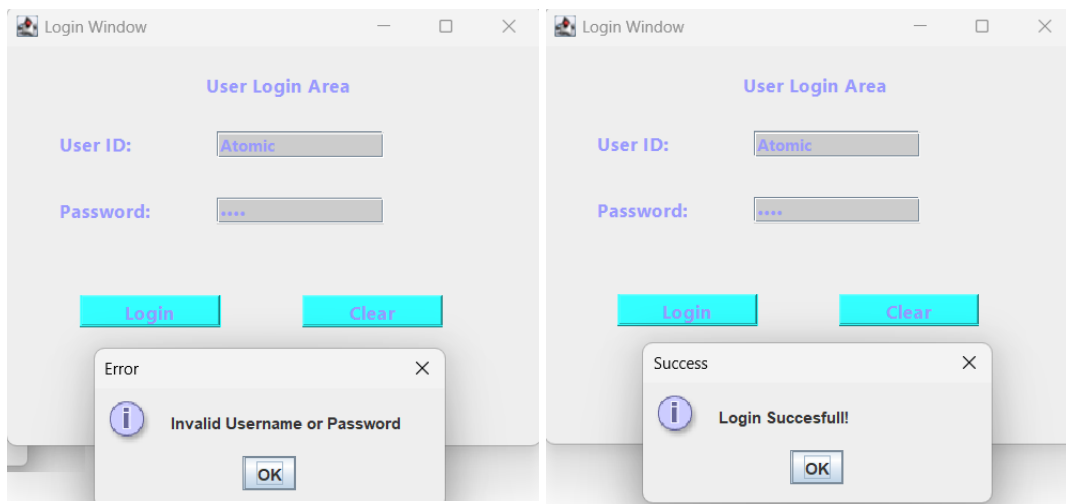
private void jButton1ActionPerformed(java.awt.event.ActionEvent evt) {
    String name = jTextField1.getText();
    String password = String.valueOf(jPasswordField1.getPassword());
    if(name.equals("Atomic") && password.equals("7878")){
        JOptionPane.showMessageDialog(this, "Login Succesfull!", "Success",
JOptionPane.INFORMATION_MESSAGE);
    }
    else{
        JOptionPane.showMessageDialog(this, "Invalid Username or Password", "Error",
JOptionPane.INFORMATION_MESSAGE);
    }
}

```

```

    }
}
private void jPasswordField1ActionPerformed(java.awt.event.ActionEvent evt) {
}
public static void main(String args[]) {
    java.awt.EventQueue.invokeLater(() -> new authentication_frame().setVisible(true));
}
// Variables declaration - do not modify
private javax.swing.JButton jButton1;
private javax.swing.JButton jButton2;
private javax.swing.JLabel jLabel1;
private javax.swing.JLabel jLabel2;
private javax.swing.JLabel jLabel3;
private javax.swing.JPasswordField jPasswordField1;
private javax.swing.JTextField jTextField1;
// End of variables declaration
}

```

OUTPUT:**3. Write a program to accept the temperature in Celsius and display Fahrenheit. (HW)****CODE:**

```

package com.mycompany.swingprojects;

public class temp_converter_frame extends javax.swing.JFrame {
    private static final java.util.logging.Logger logger = java.util.logging.Logger.getLogger(temp_converter_frame.class.getName());
    public temp_converter_frame() {
        initComponents();
    }
    @SuppressWarnings("unchecked")
    // <editor-fold defaultstate="collapsed" desc="Generated Code">
    private void initComponents() {
        jLabel1 = new javax.swing.JLabel();
        jLabel2 = new javax.swing.JLabel();
        jTextField1 = new javax.swing.JTextField();
        jTextField2 = new javax.swing.JTextField();
        jButton1 = new javax.swing.JButton();
        jLabel3 = new javax.swing.JLabel();
        setDefaultCloseOperation(javax.swing.WindowConstants.EXIT_ON_CLOSE);
        setTitle("Temperature Converter");
        setBackground(new java.awt.Color(102, 255, 102));
        jLabel1.setBackground(new java.awt.Color(255, 204, 204));
        jLabel1.setFont(new java.awt.Font("Serif", 1, 14)); // NOI18N
        jLabel1.setForeground(new java.awt.Color(0, 0, 255));
        jLabel1.setText("Temperature in  Celsius:");
        jLabel2.setBackground(new java.awt.Color(255, 204, 204));
        jLabel2.setFont(new java.awt.Font("Serif", 1, 14)); // NOI18N
        jLabel2.setForeground(new java.awt.Color(0, 0, 255));
        jLabel2.setText("Temperature in Farenheit:");
    }
}

```



```

jTextField1.setBackground(new java.awt.Color(255, 0, 0));
jTextField1.setFont(new java.awt.Font("Segoe UI", 1, 12)); // NOI18N
jTextField1.setBorder(javax.swing.BorderFactory.createEtchedBorder());
jTextField1.addActionListener(new java.awt.event.ActionListener() {
    public void actionPerformed(java.awt.event.ActionEvent evt) {
        jTextField1ActionPerformed(evt);
    }
});
jTextField2.setBackground(new java.awt.Color(255, 0, 0));
jTextField2.setFont(new java.awt.Font("Segoe UI", 1, 12)); // NOI18N
jTextField2.setBorder(javax.swing.BorderFactory.createEtchedBorder());
jButton1.setBackground(new java.awt.Color(255, 51, 51));
jButton1.setFont(new java.awt.Font("Segoe UI", 1, 14)); // NOI18N
jButton1.setText("Convert");
jButton1.setBorder(javax.swing.BorderFactory.createBevelBorder(javax.swing.border.BevelBorder.RAISED));
jButton1.addActionListener(new java.awt.event.ActionListener() {
    public void actionPerformed(java.awt.event.ActionEvent evt) {
        jButton1ActionPerformed(evt);
    }
});
jLabel3.setBackground(new java.awt.Color(255, 204, 204));
jLabel3.setFont(new java.awt.Font("Serif", 1, 14)); // NOI18N
jLabel3.setForeground(new java.awt.Color(0, 0, 255));
jLabel3.setText("Temperature Converter");
javax.swing.GroupLayout layout = new javax.swing.GroupLayout(getContentPane());
getContentPane().setLayout(layout);
layout.setHorizontalGroup(
    layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)
        .addGroup(javax.swing.GroupLayout.Alignment.TRAILING, layout.createSequentialGroup()
            .addContainerGap(javax.swing.GroupLayout.DEFAULT_SIZE, Short.MAX_VALUE)
            .addComponent(jLabel3)
            .addGap(125, 125, 125)
            .addGroup(layout.createSequentialGroup()
                .addGroup(layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)
                    .addGroup(layout.createSequentialGroup()
                        .addGroup(layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)
                            .add(layout.createSequentialGroup()
                                .addGroup(layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)
                                    .add(layout.createSequentialGroup()
                                        .addGap(45, 45, 45)
                                        .addGroup(layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING,
false)
                                            .addComponent(jLabel2, javax.swing.GroupLayout.DEFAULT_SIZE, javax.swing.GroupLayout.DEFAULT_SIZE,
Short.MAX_VALUE)
                                            .addComponent(jLabel1, javax.swing.GroupLayout.DEFAULT_SIZE, javax.swing.GroupLayout.DEFAULT_SIZE,
Short.MAX_VALUE))
                                    .addGap(33, 33, 33)
                                    .addGroup(layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING, false)
                                        .addComponent(jTextField1, javax.swing.GroupLayout.DEFAULT_SIZE, 100, Short.MAX_VALUE)
                                        .addComponent(jTextField2)))
                                .addGroup(layout.createSequentialGroup()
                                    .addGap(160, 160, 160)
                                    .addComponent(jButton1, javax.swing.GroupLayout.PREFERRED_SIZE, 71,
javax.swing.GroupLayout.PREFERRED_SIZE)))
                            .addContainerGap(60, Short.MAX_VALUE))
                        .addGap(24, 24, 24)
                        .addComponent(jLabel3, javax.swing.GroupLayout.PREFERRED_SIZE, 38, javax.swing.GroupLayout.PREFERRED_SIZE)
                        .addGap(18, 18, 18)
                        .addGroup(layout.createParallelGroup(javax.swing.GroupLayout.Alignment.BASELINE)
                            .addComponent(jLabel1)
                            .addComponent(jTextField1, javax.swing.GroupLayout.PREFERRED_SIZE, javax.swing.GroupLayout.DEFAULT_SIZE,
javax.swing.GroupLayout.PREFERRED_SIZE))
                        .addGap(42, 42, 42)
                        .addGroup(layout.createParallelGroup(javax.swing.GroupLayout.Alignment.BASELINE)
                            .addComponent(jLabel2)
                            .addComponent(jTextField2, javax.swing.GroupLayout.PREFERRED_SIZE, javax.swing.GroupLayout.DEFAULT_SIZE,
javax.swing.GroupLayout.PREFERRED_SIZE))

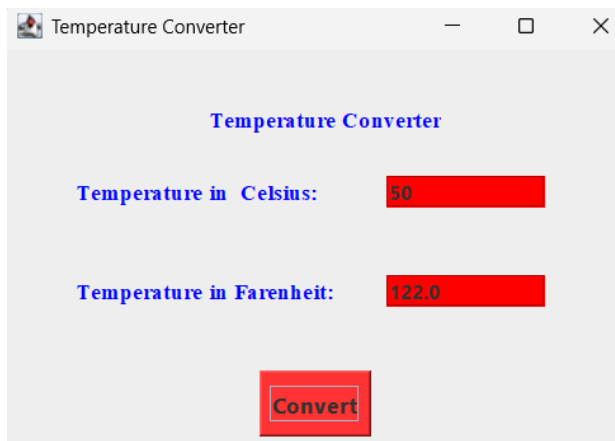
```

```

        .addGap(40, 40, 40)
        .addComponent(jButton1, javax.swing.GroupLayout.PREFERRED_SIZE, 42, javax.swing.GroupLayout.PREFERRED_SIZE)
        .addContainerGap(56, Short.MAX_VALUE)
    );

    pack();
}
private void jTextField1ActionPerformed(java.awt.event.ActionEvent evt) {}
private void jButton1ActionPerformed(java.awt.event.ActionEvent evt) {
    double temp = Double.parseDouble(jTextField1.getText());
    double res = (temp*9.0/5.0)+32;
    jTextField2.setText(String.valueOf(res));
}
public static void main(String args[]) {
    java.awt.EventQueue.invokeLater(() -> new temp_converter_frame().setVisible(true));
}
// Variables declaration - do not modify
private javax.swing.JButton jButton1;
private javax.swing.JLabel jLabel1;
private javax.swing.JLabel jLabel2;
private javax.swing.JLabel jLabel3;
private javax.swing.JTextField jTextField1;
private javax.swing.JTextField jTextField2;
// End of variables declaration
}

```

OUTPUT:**4. Write a java program to obtain the following output. (HW)****CODE:**

```

package com.mycompany.swingprojects;

import java.awt.Color;

public class feet_to_meter_frame extends javax.swing.JFrame {
    private static final java.util.logging.Logger logger = java.util.logging.Logger.getLogger(feet_to_meter_frame.class.getName());
    public feet_to_meter_frame() {
        initComponents();
        getContentPane().setBackground(new Color(102,102,255));
    }
    @SuppressWarnings("unchecked")
    // <editor-fold defaultstate="collapsed" desc="Generated Code">
    private void initComponents() {
        jLabel1 = new javax.swing.JLabel();
        jLabel2 = new javax.swing.JLabel();
        jLabel3 = new javax.swing.JLabel();
        jTextField1 = new javax.swing.JTextField();
        jButton1 = new javax.swing.JButton();
        jLabel4 = new javax.swing.JLabel();
        setDefaultCloseOperation(javax.swing.WindowConstants.EXIT_ON_CLOSE);
    }
}

```

```

setTitle("Feet to Meter Converter");
setBackground(new java.awt.Color(102, 102, 255));
jLabel1.setBackground(new java.awt.Color(255, 255, 255));
jLabel1.setFont(new java.awt.Font("Segoe UI", 1, 14)); // NOI18N
jLabel1.setForeground(new java.awt.Color(51, 255, 255));
jLabel1.setText("Feet:");
jLabel2.setBackground(new java.awt.Color(255, 255, 255));
jLabel2.setFont(new java.awt.Font("Segoe UI", 1, 14)); // NOI18N
jLabel2.setForeground(new java.awt.Color(51, 255, 255));
jLabel2.setText("Meters:");
jLabel3.setBackground(new java.awt.Color(255, 255, 255));
jLabel3.setFont(new java.awt.Font("Segoe UI", 1, 14)); // NOI18N
jLabel3.setForeground(new java.awt.Color(51, 255, 255));
jLabel3.setText("Converter");
jTextField1.setBackground(new java.awt.Color(153, 153, 153));
jTextField1.setFont(new java.awt.Font("Segoe UI", 1, 14)); // NOI18N
jTextField1.setForeground(new java.awt.Color(51, 255, 255));
jButton1.setBackground(new java.awt.Color(153, 153, 153));
jButton1.setFont(new java.awt.Font("Segoe UI", 1, 14)); // NOI18N
jButton1.setForeground(new java.awt.Color(102, 102, 255));
jButton1.setText("Convert");
jButton1.setBorder(javax.swing.BorderFactory.createLineBorder(new java.awt.Color(0, 0, 0)));
jButton1.addActionListener(new java.awt.event.ActionListener() {
    public void actionPerformed(java.awt.event.ActionEvent evt) {
        jButton1ActionPerformed(evt);
    }
});

jLabel4.setBackground(new java.awt.Color(153, 153, 153));
jLabel4.setFont(new java.awt.Font("Segoe UI", 1, 14)); // NOI18N
jLabel4.setForeground(new java.awt.Color(153, 153, 255));
javax.swing.GroupLayout layout = new javax.swing.GroupLayout(getContentPane());
getContentPane().setLayout(layout);
layout.setHorizontalGroup(
    layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)
        .addGroup(layout.createSequentialGroup()
            .add(layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)
                .addGroup(layout.createSequentialGroup()
                    .add(jLabel2, javax.swing.GroupLayout.DEFAULT_SIZE, 70, Short.MAX_VALUE)
                    .add(jLabel1, javax.swing.GroupLayout.DEFAULT_SIZE, javax.swing.GroupLayout.DEFAULT_SIZE, Short.MAX_VALUE))
                .addGroup(layout.createSequentialGroup()
                    .add(jTextField1, javax.swing.GroupLayout.DEFAULT_SIZE, 119, Short.MAX_VALUE)
                    .add(jLabel4, javax.swing.GroupLayout.DEFAULT_SIZE, javax.swing.GroupLayout.DEFAULT_SIZE, Short.MAX_VALUE)))
            .add(jButton1, javax.swing.GroupLayout.PREFERRED_SIZE, 69, javax.swing.GroupLayout.PREFERRED_SIZE)
            .add(jLabel3, javax.swing.GroupLayout.PREFERRED_SIZE, 26, javax.swing.GroupLayout.PREFERRED_SIZE)
            .addContainerGap(112, true))
        .addGroup(layout.createSequentialGroup()
            .add(jButton1, javax.swing.GroupLayout.PREFERRED_SIZE, 69, javax.swing.GroupLayout.PREFERRED_SIZE)
            .add(jLabel3, javax.swing.GroupLayout.PREFERRED_SIZE, 26, javax.swing.GroupLayout.PREFERRED_SIZE)
            .add(jTextField1, javax.swing.GroupLayout.PREFERRED_SIZE, 119, Short.MAX_VALUE)
            .add(jLabel4, javax.swing.GroupLayout.DEFAULT_SIZE, javax.swing.GroupLayout.DEFAULT_SIZE, Short.MAX_VALUE)
            .add(jLabel1, javax.swing.GroupLayout.DEFAULT_SIZE, javax.swing.GroupLayout.DEFAULT_SIZE, Short.MAX_VALUE)
            .add(jLabel2, javax.swing.GroupLayout.DEFAULT_SIZE, 70, Short.MAX_VALUE)
            .addContainerGap(112, true))
);
layout.setVerticalGroup(
    layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)
        .addGroup(layout.createSequentialGroup()
            .add(jLabel1, javax.swing.GroupLayout.PREFERRED_SIZE, 26, javax.swing.GroupLayout.PREFERRED_SIZE)
            .add(jLabel2, javax.swing.GroupLayout.PREFERRED_SIZE, 26, javax.swing.GroupLayout.PREFERRED_SIZE)
            .add(jLabel3, javax.swing.GroupLayout.PREFERRED_SIZE, 26, javax.swing.GroupLayout.PREFERRED_SIZE)
            .add(jButton1, javax.swing.GroupLayout.PREFERRED_SIZE, 26, javax.swing.GroupLayout.PREFERRED_SIZE)
            .add(jTextField1, javax.swing.GroupLayout.PREFERRED_SIZE, 26, javax.swing.GroupLayout.PREFERRED_SIZE)
            .add(jLabel4, javax.swing.GroupLayout.PREFERRED_SIZE, 26, javax.swing.GroupLayout.PREFERRED_SIZE)
            .addContainerGap(112, true))
);

```

```

        .addGap(18, 18, 18)
        .addGroup(layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING, false)
            .addComponent(jLabel2, javax.swing.GroupLayout.DEFAULT_SIZE, javax.swing.GroupLayout.DEFAULT_SIZE,
Short.MAX_VALUE)
            .addComponent(jLabel4, javax.swing.GroupLayout.DEFAULT_SIZE, javax.swing.GroupLayout.DEFAULT_SIZE,
Short.MAX_VALUE))
        .addPreferredGap(javax.swing.LayoutStyle.ComponentPlacement.RELATED, 63, Short.MAX_VALUE)
        .addComponent(jButton1, javax.swing.GroupLayout.PREFERRED_SIZE, 33, javax.swing.GroupLayout.PREFERRED_SIZE)
        .addGap(54, 54, 54))
    );
    pack();
}

private void jButton1ActionPerformed(java.awt.event.ActionEvent evt) {
    double feet = Double.parseDouble(jTextField1.getText());
    double r = feet*0.3048;
    jLabel4.setText("Equivalent to " + r);
}

public static void main(String args[]) {
    java.awt.EventQueue.invokeLater(() -> new feet_to_meter_frame().setVisible(true));
}

// Variables declaration - do not modify
private javax.swing.JButton jButton1;
private javax.swing.JLabel jLabel1;
private javax.swing.JLabel jLabel2;
private javax.swing.JLabel jLabel3;
private javax.swing.JLabel jLabel4;
private javax.swing.JTextField jTextField1;
// End of variables declaration
}

```

OUTPUT:

5. Write a program to accept information of customer for an investment policy such as Name, Date of Birth and Address. The amount to be invested should be selected from a combo box. A customer can invest from min. age of 21 to 60. According to Date of birth, determine current age of customer, total number of years remaining and accordingly find & display the premium that customer will have to pay. (H.W)

CODE:

```

package com.mycompany.swingprojects;
import java.awt.Color;
import java.time.LocalDate;
import java.time.Period;
public class customer_policy_frame extends javax.swing.JFrame {

```

```

private static final java.util.logging.Logger logger = java.util.logging.Logger.getLogger(customer_policy_frame.class.getName());
public customer_policy_frame() {
    getContentPane().setBackground(new Color(102,102,102));
    initComponents();
}
@SuppressWarnings("unchecked")
// <editor-fold defaultstate="collapsed" desc="Generated Code">
private void initComponents() {
    jLabel1 = new javax.swing.JLabel();
    jLabel2 = new javax.swing.JLabel();
    jLabel3 = new javax.swing.JLabel();
    jLabel4 = new javax.swing.JLabel();
    jLabel5 = new javax.swing.JLabel();
    jTextField1 = new javax.swing.JTextField();
    jTextField2 = new javax.swing.JTextField();
    jScrollPane1 = new javax.swing.JScrollPane();
    jTextArea1 = new javax.swing.JTextArea();
    jComboBox1 = new javax.swing.JComboBox<>();
    jButton1 = new javax.swing.JButton();
    jScrollPane2 = new javax.swing.JScrollPane();
    jTextArea2 = new javax.swing.JTextArea();
    setDefaultCloseOperation(javax.swing.WindowConstants.EXIT_ON_CLOSE);
    setTitle("Investment Window");
    setBackground(new java.awt.Color(102, 102, 102));
    jLabel1.setFont(new java.awt.Font("Viner Hand ITC", 1, 14)); // NOI18N
    jLabel1.setForeground(new java.awt.Color(51, 255, 51));
    jLabel1.setText("Investment Forum");
    jLabel2.setFont(new java.awt.Font("Viner Hand ITC", 1, 14)); // NOI18N
    jLabel2.setForeground(new java.awt.Color(51, 255, 51));
    jLabel2.setText("Name:");
    jLabel3.setFont(new java.awt.Font("Viner Hand ITC", 1, 14)); // NOI18N
    jLabel3.setForeground(new java.awt.Color(51, 255, 51));
    jLabel3.setText("Date Of Birth:");
    jLabel4.setFont(new java.awt.Font("Viner Hand ITC", 1, 14)); // NOI18N
    jLabel4.setForeground(new java.awt.Color(51, 255, 51));
    jLabel4.setText("Address:");
    jLabel5.setFont(new java.awt.Font("Viner Hand ITC", 1, 14)); // NOI18N
    jLabel5.setForeground(new java.awt.Color(51, 255, 51));
    jLabel5.setText("Investment Amount:");
    jTextField1.setBackground(new java.awt.Color(102, 102, 102));
    jTextField1.setFont(new java.awt.Font("Tw Cen MT Condensed", 1, 12)); // NOI18N
    jTextField1.setForeground(new java.awt.Color(153, 255, 153));
    jTextField1.setBorder(javax.swing.BorderFactory.createEtchedBorder());
    jTextField1.addActionListener(new java.awt.event.ActionListener() {
        public void actionPerformed(java.awt.event.ActionEvent evt) {
            jTextField1ActionPerformed(evt);
        }
    });
    jTextField2.setBackground(new java.awt.Color(102, 102, 102));
    jTextField2.setFont(new java.awt.Font("Tw Cen MT Condensed", 1, 12)); // NOI18N
    jTextField2.setForeground(new java.awt.Color(153, 255, 153));
    jTextField2.setText("yyyy-dd-mm");
    jScrollPane1.setBackground(new java.awt.Color(102, 102, 102));
    jScrollPane1.setForeground(new java.awt.Color(153, 255, 153));
    jTextArea1.setBackground(new java.awt.Color(102, 102, 102));
    jTextArea1.setColumns(20);
    jTextArea1.setFont(new java.awt.Font("Tw Cen MT Condensed", 1, 12)); // NOI18N
    jTextArea1.setForeground(new java.awt.Color(153, 255, 153));
    jTextArea1.setRows(5);
    jScrollPane1.setViewportView(jTextArea1);
    jComboBox1.setBackground(new java.awt.Color(102, 102, 102));
    jComboBox1.setFont(new java.awt.Font("Tw Cen MT Condensed", 1, 12)); // NOI18N
    jComboBox1.setForeground(new java.awt.Color(153, 255, 153));
    jComboBox1.setModel(new javax.swing.DefaultComboBoxModel<>(new String[] { "Choose Amount", "10000", "20000", "30000",
" " }));
}

```

```

jComboBox1.addActionListener(new java.awt.event.ActionListener() {
    public void actionPerformed(java.awt.event.ActionEvent evt) {
        jComboBox1ActionPerformed(evt);
    }
});
jButton1.setBackground(new java.awt.Color(102, 102, 102));
jButton1.setFont(new java.awt.Font("Tw Cen MT Condensed", 1, 12)); // NOI18N
jButton1.setForeground(new java.awt.Color(153, 255, 153));
jButton1.setText("SUBMIT");
jButton1.addActionListener(new java.awt.event.ActionListener() {
    public void actionPerformed(java.awt.event.ActionEvent evt) {
        jButton1ActionPerformed(evt);
    }
});
jScrollPane2.setBackground(new java.awt.Color(102, 102, 102));
jScrollPane2.setForeground(new java.awt.Color(153, 255, 153));

jTextArea2.setBackground(new java.awt.Color(102, 102, 102));
jTextArea2.setColumns(20);
jTextArea2.setFont(new java.awt.Font("Tw Cen MT Condensed", 1, 12)); // NOI18N
jTextArea2.setForeground(new java.awt.Color(102, 255, 102));
jTextArea2.setRows(5);
jScrollPane2.setViewportView(jTextArea2);
javax.swing.GroupLayout layout = new javax.swing.GroupLayout(getContentPane());
getContentPane().setLayout(layout);
layout.setHorizontalGroup(
    layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)
        .addGroup(layout.createSequentialGroup()
            .addGap(34, 34, 34)
            .addGroup(layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING, false)
                .addGroup(layout.createSequentialGroup()
                    .addComponent(jLabel4, javax.swing.GroupLayout.DEFAULT_SIZE, javax.swing.GroupLayout.DEFAULT_SIZE, Short.MAX_VALUE)
                    .addComponent(jLabel2, javax.swing.GroupLayout.DEFAULT_SIZE, javax.swing.GroupLayout.DEFAULT_SIZE, Short.MAX_VALUE)
                    .addComponent(jLabel5, javax.swing.GroupLayout.DEFAULT_SIZE, 174, Short.MAX_VALUE)
                    .addComponent(jLabel3, javax.swing.GroupLayout.DEFAULT_SIZE, javax.swing.GroupLayout.DEFAULT_SIZE, Short.MAX_VALUE)
                    .addPreferredGap(javax.swing.LayoutStyle.ComponentPlacement.RELATED)
                    .addGroup(layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)
                        .addGroup(layout.createSequentialGroup()
                            .addComponent(jTextField2, javax.swing.GroupLayout.Alignment.LEADING)
                            .addComponent(jScrollPane1, javax.swing.GroupLayout.Alignment.LEADING, javax.swing.GroupLayout.DEFAULT_SIZE, 170, Short.MAX_VALUE)
                            .addComponent(jTextField1))
                        .addGroup(layout.createSequentialGroup()
                            .addComponent(jComboBox1, javax.swing.GroupLayout.PREFERRED_SIZE, javax.swing.GroupLayout.DEFAULT_SIZE, javax.swing.GroupLayout.PREFERRED_SIZE)
                            .addPreferredGap(javax.swing.LayoutStyle.ComponentPlacement.RELATED, javax.swing.GroupLayout.DEFAULT_SIZE, Short.MAX_VALUE)
                            .addComponent(jButton1)))
                    .addComponent(jScrollPane2, javax.swing.GroupLayout.PREFERRED_SIZE, 365, javax.swing.GroupLayout.PREFERRED_SIZE)
                    .addGap(0, 34, Short.MAX_VALUE))
            .addGroup(layout.createSequentialGroup()
                .addGap(144, 144, 144)
                .addComponent(jLabel1)
                .addGap(20, 20, 20))
        );
layout.setVerticalGroup(
    layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)
        .addGroup(layout.createSequentialGroup()
            .addComponent(jLabel1, javax.swing.GroupLayout.PREFERRED_SIZE, 35, javax.swing.GroupLayout.PREFERRED_SIZE)
            .addPreferredGap(javax.swing.LayoutStyle.ComponentPlacement.UNRELATED)

```

```

        .addGroup(layout.createParallelGroup(javax.swing.GroupLayout.Alignment.BASELINE)
            .addComponent(jLabel2)
            .addComponent(jTextField1, javax.swing.GroupLayout.PREFERRED_SIZE, javax.swing.GroupLayout.DEFAULT_SIZE,
javax.swing.GroupLayout.PREFERRED_SIZE))
        .addPreferredGap(javax.swing.LayoutStyle.ComponentPlacement.UNRELATED)
        .addGroup(layout.createParallelGroup(javax.swing.GroupLayout.Alignment.BASELINE)
            .addComponent(jLabel3)
            .addComponent(jTextField2, javax.swing.GroupLayout.PREFERRED_SIZE, javax.swing.GroupLayout.DEFAULT_SIZE,
javax.swing.GroupLayout.PREFERRED_SIZE))
        .addPreferredGap(javax.swing.LayoutStyle.ComponentPlacement.RELATED)
        .addGroup(layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)
            .addComponent(jLabel4)
            .addComponent(jScrollPane1, javax.swing.GroupLayout.PREFERRED_SIZE, 69,
javax.swing.GroupLayout.PREFERRED_SIZE))
        .addGap(28, 28, 28)
        .addGroup(layout.createParallelGroup(javax.swing.GroupLayout.Alignment.BASELINE)
            .addComponent(jLabel5)
            .addComponent(jComboBox1, javax.swing.GroupLayout.PREFERRED_SIZE, javax.swing.GroupLayout.DEFAULT_SIZE,
javax.swing.GroupLayout.PREFERRED_SIZE))
        .addComponent(jButton1))
        .addPreferredGap(javax.swing.LayoutStyle.ComponentPlacement.RELATED)
        .addComponent(jScrollPane2, javax.swing.GroupLayout.PREFERRED_SIZE, 77, javax.swing.GroupLayout.PREFERRED_SIZE)
        .addContainerGap(21, Short.MAX_VALUE))
    );

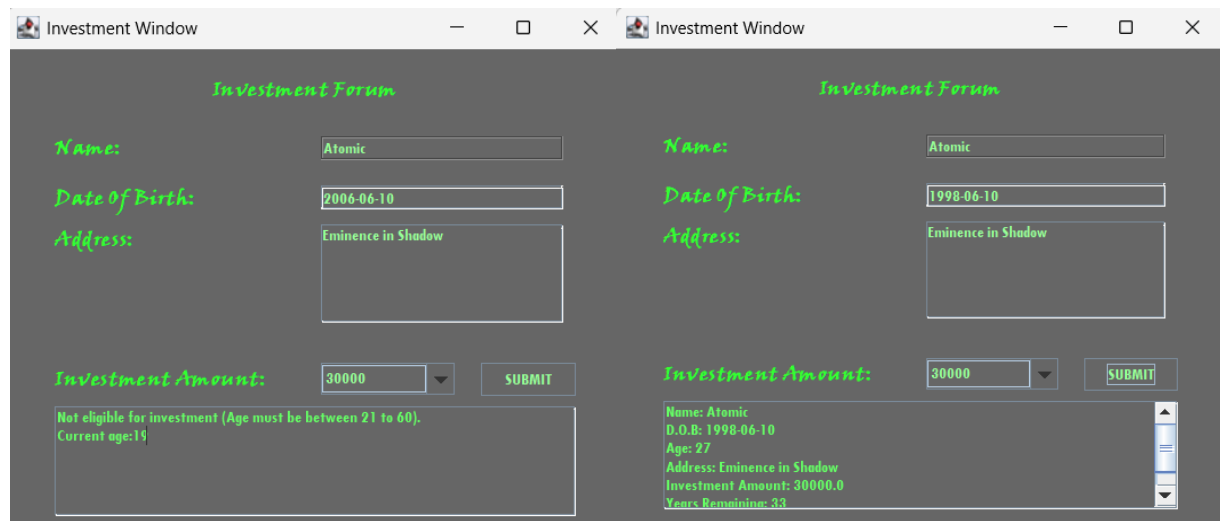
    pack();
}
private void jComboBox1ActionPerformed(java.awt.event.ActionEvent evt) {}
private void jButton1ActionPerformed(java.awt.event.ActionEvent evt) {
    try{
        String name = jTextField1.getText();
        String dob = jTextField2.getText();
        String add = jTextArea1.getText();
        double amt = Double.parseDouble(jComboBox1.getSelectedItem().toString());
        LocalDate ld = LocalDate.parse(dob);
        LocalDate td = LocalDate.now();
        int age = Period.between(ld, td).getYears();
        if(age>60 || age<21){
            jTextArea2.setText("Not eligible for investment (Age must be between 21 to
60).\nCurrent age:" + age);
            return;
        }
        int yrem = 60-age;
        double premium = amt/yrem;
        String result = "Name: " + name +
            "\nD.O.B: " + dob +
            "\nAge: " + age +
            "\nAddress: " + add +
            "\nInvestment Amount: " + amt +
            "\nYears Remaining: " + yrem +
            "\nYearly Premium: " + premium;
        jTextArea2.setText(result);
    }
    catch(Exception e){
        jTextArea2.setText("Please make sure that all fields are filled properly!!");
    }
}

```

```

    }
}
private void jTextField1ActionPerformed(java.awt.event.ActionEvent evt) {}
public static void main(String args[]) {
    java.awt.EventQueue.invokeLater(() -> new customer_policy_frame().setVisible(true));
}
// Variables declaration - do not modify
private javax.swing.JButton jButton1;
private javax.swing.JComboBox<String> jComboBox1;
private javax.swing.JLabel jLabel1;
private javax.swing.JLabel jLabel2;
private javax.swing.JLabel jLabel3;
private javax.swing.JLabel jLabel4;
private javax.swing.JLabel jLabel5;
private javax.swing.JScrollPane jScrollPane1;
private javax.swing.JScrollPane jScrollPane2;
private javax.swing.JTextArea jTextArea1;
private javax.swing.JTextArea jTextArea2;
private javax.swing.JTextField jTextField1;
private javax.swing.JTextField jTextField2;
// End of variables declaration
}

```

OUTPUT:

6. Write a java Program to create a window as described below: In the window there are four radio buttons Add, Sub, Multi, div at the top. This group is called Operation. There are two TextFields to input first and second number. We must type the number and select one of the operations. On selection of operation the result should be displayed.

CODE:

```

package com.mycompany.swingprojects;
public class group_operation_arithmetic_frame extends javax.swing.JFrame {
    private static final java.util.logging.Logger logger =
        java.util.logging.Logger.getLogger(group_operation_arithmetic_frame.class.getName());
    public group_operation_arithmetic_frame() {
        initComponents();
    }
    @SuppressWarnings("unchecked")
    // <editor-fold defaultstate="collapsed" desc="Generated Code">
    private void initComponents() {

        jRadioButton1 = new javax.swing.JRadioButton();
        jRadioButton2 = new javax.swing.JRadioButton();
        jRadioButton3 = new javax.swing.JRadioButton();
        jRadioButton4 = new javax.swing.JRadioButton();
        jLabel1 = new javax.swing.JLabel();

```



```

jLabel2 = new javax.swing.JLabel();
jTextField1 = new javax.swing.JTextField();
jTextField2 = new javax.swing.JTextField();
jButton1 = new javax.swing.JButton();
jLabel3 = new javax.swing.JLabel();
setDefaultCloseOperation(javax.swing.WindowConstants.EXIT_ON_CLOSE);
jRadioButton1.setFont(new java.awt.Font("Segoe UI", 1, 12)); // NOI18N
jRadioButton1.setForeground(new java.awt.Color(0, 0, 255));
jRadioButton1.setText("ADD");
jRadioButton1.addActionListener(new java.awt.event.ActionListener() {
    public void actionPerformed(java.awt.event.ActionEvent evt) {
        jRadioButton1ActionPerformed(evt);
    }
});
jRadioButton2.setFont(new java.awt.Font("Segoe UI", 1, 12)); // NOI18N
jRadioButton2.setForeground(new java.awt.Color(0, 0, 255));
jRadioButton2.setText("SUBTRACT");
jRadioButton2.addActionListener(new java.awt.event.ActionListener() {
    public void actionPerformed(java.awt.event.ActionEvent evt) {
        jRadioButton2ActionPerformed(evt);
    }
});
jRadioButton3.setFont(new java.awt.Font("Segoe UI", 1, 12)); // NOI18N
jRadioButton3.setForeground(new java.awt.Color(0, 0, 255));
jRadioButton3.setText("MULTIPLY");
jRadioButton4.setFont(new java.awt.Font("Segoe UI", 1, 12)); // NOI18N
jRadioButton4.setForeground(new java.awt.Color(0, 0, 255));
jRadioButton4.setText("DIVISION");
jRadioButton4.addActionListener(new java.awt.event.ActionListener() {
    public void actionPerformed(java.awt.event.ActionEvent evt) {
        jRadioButton4ActionPerformed(evt);
    }
});
jLabel1.setBackground(new java.awt.Color(102, 102, 102));
jLabel1.setFont(new java.awt.Font("Segoe UI", 1, 14)); // NOI18N
jLabel1.setForeground(new java.awt.Color(0, 0, 255));
jLabel1.setText("1st Number:");

jLabel2.setBackground(new java.awt.Color(102, 102, 102));
jLabel2.setFont(new java.awt.Font("Segoe UI", 1, 14)); // NOI18N
jLabel2.setForeground(new java.awt.Color(0, 0, 255));
jLabel2.setText("2nd Number:");
jTextField1.setBackground(new java.awt.Color(204, 204, 204));
jTextField1.setFont(new java.awt.Font("Segoe UI", 1, 12)); // NOI18N
jTextField1.setForeground(new java.awt.Color(0, 0, 255));
jTextField2.setBackground(new java.awt.Color(204, 204, 204));
jTextField2.setFont(new java.awt.Font("Segoe UI", 1, 12)); // NOI18N
jTextField2.setForeground(new java.awt.Color(0, 0, 255));
jButton1.setBackground(new java.awt.Color(204, 204, 204));
jButton1.setFont(new java.awt.Font("Segoe UI", 1, 12)); // NOI18N
jButton1.setForeground(new java.awt.Color(0, 0, 255));
jButton1.setText("Result");
jButton1.addActionListener(new java.awt.event.ActionListener() {
    public void actionPerformed(java.awt.event.ActionEvent evt) {
        jButton1ActionPerformed(evt);
    }
});
jLabel3.setBackground(new java.awt.Color(102, 102, 102));
jLabel3.setFont(new java.awt.Font("Segoe UI", 1, 14)); // NOI18N
jLabel3.setForeground(new java.awt.Color(0, 0, 255));
javax.swing.GroupLayout layout = new javax.swing.GroupLayout(getContentPane());
getContentPane().setLayout(layout);
layout.setHorizontalGroup(
    layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)
        .addGroup(layout.createSequentialGroup()
            .add(layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)
                .add(jLabel1)
                .add(jLabel2)
                .add(jTextField1)
                .add(jTextField2)
                .add(jButton1)
                .add(jLabel3)
            )
            .addGap(62, 62, 62)
        )
);

```

```

        .addGroup(layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)
            .addGroup(layout.createSequentialGroup())
        .addGroup(layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING, false)
            .addComponent(jRadioButton3, javax.swing.GroupLayout.DEFAULT_SIZE, javax.swing.GroupLayout.DEFAULT_SIZE,
Short.MAX_VALUE)
            .addComponent(jLabel1, javax.swing.GroupLayout.DEFAULT_SIZE, 98, Short.MAX_VALUE)
            .addComponent(jRadioButton1, javax.swing.GroupLayout.DEFAULT_SIZE, javax.swing.GroupLayout.DEFAULT_SIZE,
Short.MAX_VALUE)
            .addComponent(jTextField1))
            .addGap(45, 45, 45)
        .addGroup(layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING,
false)
            .addComponent(jLabel2, javax.swing.GroupLayout.DEFAULT_SIZE, 98, Short.MAX_VALUE)
            .addComponent(jRadioButton2, javax.swing.GroupLayout.DEFAULT_SIZE, javax.swing.GroupLayout.DEFAULT_SIZE,
Short.MAX_VALUE)
            .addComponent(jRadioButton4, javax.swing.GroupLayout.DEFAULT_SIZE, javax.swing.GroupLayout.DEFAULT_SIZE,
Short.MAX_VALUE)
            .addComponent(jTextField2)))
        .addGroup(layout.createSequentialGroup())
        .addComponent(jButton1)
        .addPreferredGap(javax.swing.LayoutStyle.ComponentPlacement.RELATED)
        .addComponent(jLabel3, javax.swing.GroupLayout.PREFERRED_SIZE, 230,
javax.swing.GroupLayout.PREFERRED_SIZE))
        .addContainerGap(30, Short.MAX_VALUE))
    );
    layout.setVerticalGroup(
        layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)
        .addGroup(layout.createSequentialGroup())
        .addGap(37, 37, 37)
        .addGroup(layout.createParallelGroup(javax.swing.GroupLayout.Alignment.BASELINE)
            .addComponent(jLabel1)
            .addComponent(jLabel2))
        .addGap(18, 18, 18)
        .addGroup(layout.createParallelGroup(javax.swing.GroupLayout.Alignment.BASELINE)
            .addComponent(jTextField1, javax.swing.GroupLayout.PREFERRED_SIZE, javax.swing.GroupLayout.DEFAULT_SIZE,
javax.swing.GroupLayout.PREFERRED_SIZE)
            .addComponent(jTextField2, javax.swing.GroupLayout.PREFERRED_SIZE, javax.swing.GroupLayout.DEFAULT_SIZE,
javax.swing.GroupLayout.PREFERRED_SIZE))
        .addGap(39, 39, 39)
        .addGroup(layout.createParallelGroup(javax.swing.GroupLayout.Alignment.BASELINE)
            .addComponent(jRadioButton1)
            .addComponent(jRadioButton2))
        .addPreferredGap(javax.swing.LayoutStyle.ComponentPlacement.RELATED)
        .addGroup(layout.createParallelGroup(javax.swing.GroupLayout.Alignment.BASELINE)
            .addComponent(jRadioButton3)
            .addComponent(jRadioButton4))
        .addGap(45, 45, 45)
        .addGroup(layout.createParallelGroup(javax.swing.GroupLayout.Alignment.BASELINE)
            .addComponent(jButton1)
            .addComponent(jLabel3))
        .addContainerGap(48, Short.MAX_VALUE))
    );
    pack();
}

private void jRadioButton1ActionPerformed(java.awt.event.ActionEvent evt) {}
private void jRadioButton2ActionPerformed(java.awt.event.ActionEvent evt) {}
private void jRadioButton4ActionPerformed(java.awt.event.ActionEvent evt) {}
private void jButton1ActionPerformed(java.awt.event.ActionEvent evt) {
    double x = Double.parseDouble(jTextField1.getText());
    double y = Double.parseDouble(jTextField2.getText());
    double r = 0;
    try{
        if(jRadioButton1.isSelected()){
            r = x+y;

```

```
    }
    if(jRadioButton2.isSelected()){
        r = x-y;
    }
    if(jRadioButton3.isSelected()){
        r = x*y;
    }
    if(jRadioButton4.isSelected()){
        if(y==0){
            jLabel3.setText("Can't divide by Zero!");
            return;
        }
        r = x/y;
    }
    jLabel3.setText("Result: " + r);
}
catch(Exception e){
    jLabel3.setText("Enter Valid Number!!");
}
}
}
public static void main(String args[]) {
    java.awt.EventQueue.invokeLater(() -> new group_operation_arithmetic_frame().setVisible(true));
}
// Variables declaration - do not modify
private javax.swing.JButton jButton1;
private javax.swing.JLabel jLabel1;
private javax.swing.JLabel jLabel2;
private javax.swing.JLabel jLabel3;
private javax.swing.JRadioButton jRadioButton1;
private javax.swing.JRadioButton jRadioButton2;
private javax.swing.JRadioButton jRadioButton3;
private javax.swing.JRadioButton jRadioButton4;
private javax.swing.JTextField jTextField1;
private javax.swing.JTextField jTextField2;
// End of variables declaration
}
```

OUTPUT:

1st Number: 90 2nd Number: 40

☒ ADD
 ☐ SUBTRACT
☐ MULTIPLY
 ☐ DIVISION

Result Result: 130.0

1st Number: 90 2nd Number: 40

☐ ADD
 ☒ SUBTRACT
☐ MULTIPLY
 ☐ DIVISION

Result Result: 50.0

1st Number: 90 2nd Number: 40

☐ ADD
 ☐ SUBTRACT
☒ MULTIPLY
 ☐ DIVISION

Result Result: 3600.0

1st Number: 90 2nd Number: 40

☐ ADD
 ☐ SUBTRACT
☐ MULTIPLY
 ☒ DIVISION

Result Result: 2.25

7. Write a swing program to find square of given number.

CODE:

```
package com.mycompany.swingprojects;

public class square_number_frame extends javax.swing.JFrame {
    private static final java.util.logging.Logger logger = java.util.logging.Logger.getLogger(square_number_frame.class.getName());
    public square_number_frame() {
        initComponents();
    }
    @SuppressWarnings("unchecked")
    // <editor-fold defaultstate="collapsed" desc="Generated Code">
    private void initComponents() {
        jLabel1 = new javax.swing.JLabel();
        jLabel2 = new javax.swing.JLabel();
        jTextField1 = new javax.swing.JTextField();
        jButton1 = new javax.swing.JButton();
        jButton2 = new javax.swing.JButton();
        setDefaultCloseOperation(javax.swing.WindowConstants.EXIT_ON_CLOSE);
        jLabel1.setFont(new java.awt.Font("Segoe UI", 1, 14)); // NOI18N
        jLabel1.setForeground(new java.awt.Color(0, 255, 255));
        jLabel1.setText("Square Finder");
        jLabel2.setFont(new java.awt.Font("Segoe UI", 1, 14)); // NOI18N
        jLabel2.setForeground(new java.awt.Color(0, 255, 255));
        jLabel2.setText("Enter Number");
        jTextField1.setBackground(new java.awt.Color(255, 204, 204));
        jTextField1.setFont(new java.awt.Font("Segoe UI", 1, 12)); // NOI18N
        jTextField1.setForeground(new java.awt.Color(51, 204, 255));
        jTextField1.addActionListener(new java.awt.event.ActionListener() {
            public void actionPerformed(java.awt.event.ActionEvent evt) {
                jTextField1ActionPerformed(evt);
            }
        });
        jButton1.setBackground(new java.awt.Color(255, 204, 204));
```

```

jButton1.setFont(new java.awt.Font("Segoe UI", 1, 12)); // NOI18N
jButton1.setForeground(new java.awt.Color(51, 153, 255));
jButton1.setText("Calculate");
jButton1.addActionListener(new java.awt.event.ActionListener() {
    public void actionPerformed(java.awt.event.ActionEvent evt) {
        jButton1ActionPerformed(evt);
    }
});
jButton2.setBackground(new java.awt.Color(255, 204, 204));
jButton2.setFont(new java.awt.Font("Segoe UI", 1, 12)); // NOI18N
jButton2.setForeground(new java.awt.Color(102, 153, 255));

javax.swing.GroupLayout layout = new javax.swing.GroupLayout(getContentPane());
getContentPane().setLayout(layout);
layout.setHorizontalGroup(
    layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)
        .addGroup(javax.swing.GroupLayout.Alignment.TRAILING, layout.createSequentialGroup()
            .addContainerGap()
            .addComponent(jLabel1, javax.swing.GroupLayout.PREFERRED_SIZE, 109, javax.swing.GroupLayout.PREFERRED_SIZE)
            .addPreferredGap(javax.swing.LayoutStyle.ComponentPlacement.RELATED)
            .addComponent(jLabel2, javax.swing.GroupLayout.PREFERRED_SIZE, 110, javax.swing.GroupLayout.PREFERRED_SIZE)
            .addGap(133, 133, 133)
            .addGroup(layout.createSequentialGroup()
                .addComponent(jButton2, javax.swing.GroupLayout.PREFERRED_SIZE, 109, javax.swing.GroupLayout.PREFERRED_SIZE)
                .addGap(148, 148, 148)
                .addComponent(jButton1, javax.swing.GroupLayout.PREFERRED_SIZE, 109, javax.swing.GroupLayout.PREFERRED_SIZE)
                .addPreferredGap(javax.swing.LayoutStyle.ComponentPlacement.RELATED)
                .addComponent(jTextField1, javax.swing.GroupLayout.PREFERRED_SIZE, 109, javax.swing.GroupLayout.PREFERRED_SIZE)
                .addContainerGap(143, true))
        .addGroup(layout.createSequentialGroup()
            .addGap(25, 25, 25)
            .addComponent(jLabel1, javax.swing.GroupLayout.PREFERRED_SIZE, 46, javax.swing.GroupLayout.PREFERRED_SIZE)
            .addPreferredGap(javax.swing.LayoutStyle.ComponentPlacement.UNRELATED)
            .addComponent(jLabel2)
            .addPreferredGap(javax.swing.LayoutStyle.ComponentPlacement.RELATED)
            .addComponent(jButton1)
            .addPreferredGap(javax.swing.LayoutStyle.ComponentPlacement.RELATED)
            .addComponent(jButton2, javax.swing.GroupLayout.PREFERRED_SIZE, 20, javax.swing.GroupLayout.PREFERRED_SIZE)
            .addContainerGap(78, true))
);
pack();
}

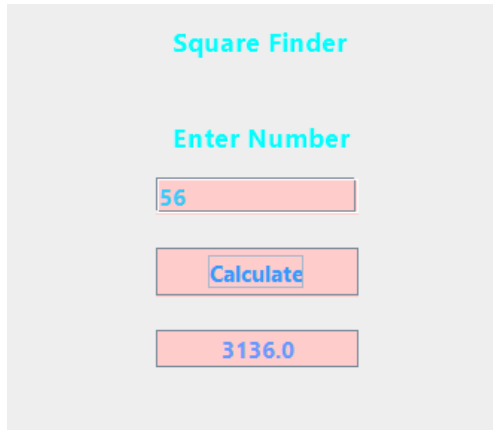
private void jButton1ActionPerformed(java.awt.event.ActionEvent evt) {
    double n = Double.parseDouble(jTextField1.getText());
    jButton2.setText(String.valueOf(n*n));
}

private void jTextField1ActionPerformed(java.awt.event.ActionEvent evt) {
}

public static void main(String args[]) {
    java.awt.EventQueue.invokeLater(() -> new square_number_frame().setVisible(true));
}

```

```
// Variables declaration - do not modify
private javax.swing.JButton jButton1;
private javax.swing.JButton jButton2;
private javax.swing.JLabel jLabel1;
private javax.swing.JLabel jLabel2;
private javax.swing.JTextField jTextField1;
// End of variables declaration
}
```

OUTPUT:

8. Write a GUI program to accept the following elements of the classes employee: i.e. Emp no, Basic pay, Dept. The salary slips along with calculation of DA, HRA and CCA should be printed on the message box. Where, DA=81% of Basic Salary if Basic Salary<5000 DA=51% of Basic Salary if Basic Salary is in the range of 5000 to 7000 DA=41% of Basic Salary if Basic Salary>7000 HRA=15% of Basic Salary CCA=350

CODE:

```
package com.mycompany.swingprojects;

import javax.swing.JOptionPane;

public class employee_salary_slip_frame extends javax.swing.JFrame {
    private static final java.util.logging.Logger logger =
        java.util.logging.Logger.getLogger(employee_salary_slip_frame.class.getName());

    public employee_salary_slip_frame() {
        initComponents();
    }

    @SuppressWarnings("unchecked")
    // <editor-fold defaultstate="collapsed" desc="Generated Code">
    private void initComponents() {
        jLabel1 = new javax.swing.JLabel();
        jLabel2 = new javax.swing.JLabel();
        jLabel3 = new javax.swing.JLabel();
        jTextField1 = new javax.swing.JTextField();
        jTextField2 = new javax.swing.JTextField();
        jTextField3 = new javax.swing.JTextField();
        jButton1 = new javax.swing.JButton();
        setDefaultCloseOperation(javax.swing.WindowConstants.EXIT_ON_CLOSE);
        jLabel1.setText("Employee ID:");

        jLabel2.setText("Basic Salary:");
        jLabel3.setText("Department:");
        jButton1.setText("Submit");
        jButton1.addActionListener(new java.awt.event.ActionListener() {
            public void actionPerformed(java.awt.event.ActionEvent evt) {
                jButton1ActionPerformed(evt);
            }
        });

        javax.swing.GroupLayout layout = new javax.swing.GroupLayout(getContentPane());
        getContentPane().setLayout(layout);
        layout.setHorizontalGroup(
```

```

        layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)
        .addGroup(layout.createSequentialGroup())
        .addContainerGap(11, Short.MAX_VALUE)
        .addGroup(layout.createParallelGroup(javax.swing.GroupLayout.Alignment.TRAILING, false)
        .addComponent(jLabel2, javax.swing.GroupLayout.Alignment.LEADING, javax.swing.GroupLayout.DEFAULT_SIZE,
javax.swing.GroupLayout.DEFAULT_SIZE, Short.MAX_VALUE)
        .addComponent(jLabel1, javax.swing.GroupLayout.Alignment.LEADING, javax.swing.GroupLayout.DEFAULT_SIZE, 105,
Short.MAX_VALUE)
        .addComponent(jLabel3, javax.swing.GroupLayout.Alignment.LEADING, javax.swing.GroupLayout.DEFAULT_SIZE,
javax.swing.GroupLayout.DEFAULT_SIZE, Short.MAX_VALUE))
        .addGap(18, 18, 18)
        .addGroup(layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)
        .addComponent(jTextField1, javax.swing.GroupLayout.PREFERRED_SIZE, 71, javax.swing.GroupLayout.PREFERRED_SIZE)
        .addComponent(jTextField2, javax.swing.GroupLayout.PREFERRED_SIZE, 71, javax.swing.GroupLayout.PREFERRED_SIZE)
        .addComponent(jTextField3, javax.swing.GroupLayout.PREFERRED_SIZE, 71,
javax.swing.GroupLayout.PREFERRED_SIZE))
        .addContainerGap(195, Short.MAX_VALUE))
        .addGroup(layout.createSequentialGroup())
        .addGap(65, 65, 65)
        .addComponent(jButton1)
        .addContainerGap(javax.swing.GroupLayout.DEFAULT_SIZE, Short.MAX_VALUE)
    );
    layout.setVerticalGroup(
        layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)
        .addGroup(layout.createSequentialGroup())
        .addGap(44, 44, 44)
        .addGroup(layout.createParallelGroup(javax.swing.GroupLayout.Alignment.BASELINE)
        .addComponent(jLabel1)
        .addComponent(jTextField1, javax.swing.GroupLayout.PREFERRED_SIZE, javax.swing.GroupLayout.DEFAULT_SIZE,
javax.swing.GroupLayout.PREFERRED_SIZE))
        .addGap(20, 20, 20)
        .addGroup(layout.createParallelGroup(javax.swing.GroupLayout.Alignment.BASELINE)
        .addComponent(jLabel2)
        .addComponent(jTextField2, javax.swing.GroupLayout.PREFERRED_SIZE, javax.swing.GroupLayout.DEFAULT_SIZE,
javax.swing.GroupLayout.PREFERRED_SIZE))
        .addGap(20, 20, 20)
        .addGroup(layout.createParallelGroup(javax.swing.GroupLayout.Alignment.BASELINE)
        .addComponent(jLabel3)
        .addComponent(jTextField3, javax.swing.GroupLayout.PREFERRED_SIZE, javax.swing.GroupLayout.DEFAULT_SIZE,
javax.swing.GroupLayout.PREFERRED_SIZE))
        .addGap(38, 38, 38)
        .addComponent(jButton1)
        .addContainerGap(89, Short.MAX_VALUE)
    );
    pack();
}

```

```

private void jButton1ActionPerformed(java.awt.event.ActionEvent evt) {

```

```

    try{

```

```

        int eno = Integer.parseInt(jTextField1.getText());

```

```

        double basic = Double.parseDouble(jTextField2.getText());

```

```

        String dep = jTextField3.getText();

```

```

        double da;

```

```

        if(basic<5000){

```

```

            da = basic*0.81;

```

```

        }

```

```

        else{

```

```

            if(basic<=7000){

```

```

                da = basic*0.51;

```

```

            }

```

```

        else{

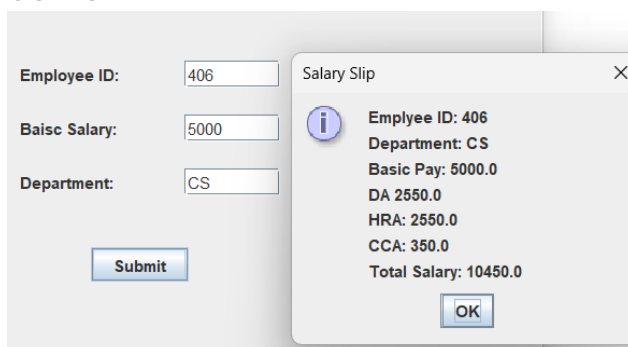
```

```

        da = basic*0.41;
    }
}
double hra = basic*0.51;
double cca = 350;
double total = basic + da + hra + cca;
String msg = "Employee ID: " + eno +
    "\nDepartment: " + dep +
    "\nBasic Pay: " + basic +
    "\nDA " + da +
    "\nHRA: " + hra +
    "\nCCA: " + cca +
    "\nTotal Salary: " + total;
JOptionPane.showMessageDialog(this, msg, "Salary Slip",
JOptionPane.INFORMATION_MESSAGE);
}
catch(Exception e){
    JOptionPane.showMessageDialog(this, "Please enter valid numbers for Basic pay and
Employee ID");
}
}
}
public static void main(String args[]) {
    java.awt.EventQueue.invokeLater(() -> new employee_salary_slip_frame().setVisible(true));
}

// Variables declaration - do not modify
private javax.swing.JButton jButton1;
private javax.swing.JLabel jLabel1;
private javax.swing.JLabel jLabel2;
private javax.swing.JLabel jLabel3;
private javax.swing.JTextField jTextField1;
private javax.swing.JTextField jTextField2;
private javax.swing.JTextField jTextField3;
// End of variables declaration
}

```

OUTPUT:

9. Write a java program to display the following GUI. On submission, all data should be displayed in message box.

CODE:

```

package com.mycompany.swingprojects;

import javax.swing.JOptionPane;

public class message_box_frame extends javax.swing.JFrame {
    private static final java.util.logging.Logger logger = java.util.logging.Logger.getLogger(message_box_frame.class.getName());
    public message_box_frame() {

```



```

initComponents();
}
@SuppressWarnings("unchecked")
// <editor-fold defaultstate="collapsed" desc="Generated Code">
private void initComponents() {
    jLabel1 = new javax.swing.JLabel();
    jLabel2 = new javax.swing.JLabel();
    jLabel3 = new javax.swing.JLabel();
    jLabel4 = new javax.swing.JLabel();
    jLabel5 = new javax.swing.JLabel();
    jLabel6 = new javax.swing.JLabel();
    jLabel7 = new javax.swing.JLabel();
    jTextField1 = new javax.swing.JTextField();
    jTextField2 = new javax.swing.JTextField();
    jScrollPane1 = new javax.swing.JScrollPane();
    jTextArea1 = new javax.swing.JTextArea();
    jTextField3 = new javax.swing.JTextField();
    jRadioButton1 = new javax.swing.JRadioButton();
    jRadioButton2 = new javax.swing.JRadioButton();
    jComboBox1 = new javax.swing.JComboBox<>();
    jCheckBox1 = new javax.swing.JCheckBox();
    jCheckBox2 = new javax.swing.JCheckBox();
    jCheckBox3 = new javax.swing.JCheckBox();
    jCheckBox4 = new javax.swing.JCheckBox();
    jButton1 = new javax.swing.JButton();
    setDefaultCloseOperation(javax.swing.WindowConstants.EXIT_ON_CLOSE);
    jLabel1.setFont(new java.awt.Font("Tempus Sans ITC", 1, 14)); // NOI18N
    jLabel1.setForeground(new java.awt.Color(0, 144, 118));
    jLabel1.setText("Name:");
    jLabel2.setFont(new java.awt.Font("Tempus Sans ITC", 1, 14)); // NOI18N
    jLabel2.setForeground(new java.awt.Color(0, 144, 118));
    jLabel2.setText("Age:");
    jLabel3.setFont(new java.awt.Font("Tempus Sans ITC", 1, 14)); // NOI18N
    jLabel3.setForeground(new java.awt.Color(0, 144, 118));
    jLabel3.setText("Address:");
    jLabel4.setFont(new java.awt.Font("Tempus Sans ITC", 1, 14)); // NOI18N
    jLabel4.setForeground(new java.awt.Color(0, 144, 118));
    jLabel4.setText("Phone No:");
    jLabel5.setFont(new java.awt.Font("Tempus Sans ITC", 1, 14)); // NOI18N
    jLabel5.setForeground(new java.awt.Color(0, 144, 118));
    jLabel5.setText("Gender:");
    jLabel6.setFont(new java.awt.Font("Tempus Sans ITC", 1, 14)); // NOI18N
    jLabel6.setForeground(new java.awt.Color(0, 144, 118));
    jLabel6.setText("Class:");
    jLabel7.setFont(new java.awt.Font("Tempus Sans ITC", 1, 14)); // NOI18N
    jLabel7.setForeground(new java.awt.Color(0, 144, 118));
    jLabel7.setText("Interpreted Language:");
    jTextArea1.setColumns(20);
    jTextArea1.setRows(5);
    jScrollPane1.setViewportView(jTextArea1);
    jRadioButton1.setText("Male");
    jRadioButton1.addActionListener(new java.awt.event.ActionListener() {
        public void actionPerformed(java.awt.event.ActionEvent evt) {
            jRadioButton1ActionPerformed(evt);
        }
    });
    jRadioButton2.setText("Female");
    jRadioButton2.addActionListener(new java.awt.event.ActionListener() {
        public void actionPerformed(java.awt.event.ActionEvent evt) {
            jRadioButton2ActionPerformed(evt);
        }
    });
    jComboBox1.setModel(new javax.swing.DefaultComboBoxModel<>(new String[] { "Choose.....", "FYCS", "SYCS", "TYCS", "FYIT",
"SYIT", "TYIT" }));
    jCheckBox1.setText("C++");
    jCheckBox1.addActionListener(new java.awt.event.ActionListener() {

```



```

        .addGap(18, 18, 18)
        .addComponent(jRadioButton2, javax.swing.GroupLayout.PREFERRED_SIZE, 98,
javax.swing.GroupLayout.PREFERRED_SIZE)))
        .addGroup(layout.createSequentialGroup())
        .addGap(168, 168, 168)
        .addComponent(jButton1)))
        .addContainerGap(javax.swing.GroupLayout.DEFAULT_SIZE, Short.MAX_VALUE)
    );
    layout.setVerticalGroup(
        layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)
        .addGroup(layout.createSequentialGroup())
        .addGap(22, 22, 22)
        .addGroup(layout.createParallelGroup(javax.swing.GroupLayout.Alignment.BASELINE)
        .addComponent(jLabel1)
        .addComponent(jTextField1, javax.swing.GroupLayout.PREFERRED_SIZE, javax.swing.GroupLayout.DEFAULT_SIZE,
javax.swing.GroupLayout.PREFERRED_SIZE))
        .addPreferredGap(javax.swing.LayoutStyle.ComponentPlacement.RELATED)
        .addGroup(layout.createParallelGroup(javax.swing.GroupLayout.Alignment.BASELINE)
        .addComponent(jLabel2)
        .addComponent(jTextField2, javax.swing.GroupLayout.PREFERRED_SIZE, javax.swing.GroupLayout.DEFAULT_SIZE,
javax.swing.GroupLayout.PREFERRED_SIZE))
        .addGap(11, 11, 11)
        .addGroup(layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)
        .addComponent(jScrollPane1, javax.swing.GroupLayout.PREFERRED_SIZE, 47,
javax.swing.GroupLayout.PREFERRED_SIZE)
        .addComponent(jLabel3))
        .addPreferredGap(javax.swing.LayoutStyle.ComponentPlacement.RELATED)
        .addGroup(layout.createParallelGroup(javax.swing.GroupLayout.Alignment.BASELINE)
        .addComponent(jTextField3, javax.swing.GroupLayout.PREFERRED_SIZE, javax.swing.GroupLayout.DEFAULT_SIZE,
javax.swing.GroupLayout.PREFERRED_SIZE)
        .addComponent(jLabel4))
        .addPreferredGap(javax.swing.LayoutStyle.ComponentPlacement.RELATED, javax.swing.GroupLayout.DEFAULT_SIZE,
Short.MAX_VALUE)
        .addGroup(layout.createParallelGroup(javax.swing.GroupLayout.Alignment.BASELINE)
        .addComponent(jLabel5)
        .addComponent(jRadioButton1)
        .addComponent(jRadioButton2))
        .addPreferredGap(javax.swing.LayoutStyle.ComponentPlacement.RELATED)
        .addGroup(layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)
        .addComponent(jLabel6)
        .addComponent(jComboBox1, javax.swing.GroupLayout.PREFERRED_SIZE, javax.swing.GroupLayout.DEFAULT_SIZE,
javax.swing.GroupLayout.PREFERRED_SIZE))
        .addGap(18, 18, 18)
        .addGroup(layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)
        .addComponent(jLabel7)
        .addGroup(layout.createParallelGroup(javax.swing.GroupLayout.Alignment.BASELINE)
        .addComponent(jCheckBox1)
        .addComponent(jCheckBox2)))
        .addPreferredGap(javax.swing.LayoutStyle.ComponentPlacement.UNRELATED)
        .addGroup(layout.createParallelGroup(javax.swing.GroupLayout.Alignment.BASELINE)
        .addComponent(jCheckBox3)
        .addComponent(jCheckBox4))
        .addGap(18, 18, 18)
        .addComponent(jButton1)
        .addGap(41, 41, 41))
    );
    pack();
}

private void jRadioButton1ActionPerformed(java.awt.event.ActionEvent evt) {}
private void jRadioButton2ActionPerformed(java.awt.event.ActionEvent evt) {}
private void jCheckBox2ActionPerformed(java.awt.event.ActionEvent evt) {}
private void jCheckBox1ActionPerformed(java.awt.event.ActionEvent evt) {}
private void jButton1ActionPerformed(java.awt.event.ActionEvent evt) {
    String name = jTextField1.getText();
    String age = jTextField2.getText();

```

```

String pno = jTextField3.getText();
String add = jTextArea1.getText();
String gender = "";
if(jRadioButton1.isSelected()){
    gender = "Male";
}
else{
    gender = "Female";
}
String sclass = (String) jComboBox1.getSelectedItem();
String lang = "Languages Known:";
if(jCheckBox1.isSelected()) lang+="C++";
if(jCheckBox2.isSelected()) lang+="PYTHON";
if(jCheckBox3.isSelected()) lang+="JAVA";
if(jCheckBox4.isSelected()) lang+="C";
String msg = "Name: " + name +
    "\nAge: " + age +
    "\nPhone No: " + pno +
    "\nAddress: " + add +
    "\nClass: " + sclass +
    "\n" + lang;
JOptionPane.showMessageDialog(this, msg, "Student details",
JOptionPane.INFORMATION_MESSAGE);
}

public static void main(String args[]) {
    java.awt.EventQueue.invokeLater(() -> new message_box_frame().setVisible(true));
}

// Variables declaration - do not modify
private javax.swing.JButton jButton1;
private javax.swing.JCheckBox jCheckBox1;
private javax.swing.JCheckBox jCheckBox2;
private javax.swing.JCheckBox jCheckBox3;
private javax.swing.JCheckBox jCheckBox4;
private javax.swing.JComboBox<String> jComboBox1;
private javax.swing.JLabel jLabel1;
private javax.swing.JLabel jLabel2;
private javax.swing.JLabel jLabel3;
private javax.swing.JLabel jLabel4;
private javax.swing.JLabel jLabel5;
private javax.swing.JLabel jLabel6;
private javax.swing.JLabel jLabel7;
private javax.swing.JRadioButton jRadioButton1;
private javax.swing.JRadioButton jRadioButton2;
private javax.swing.JScrollPane jScrollPane1;
private javax.swing.JTextArea jTextArea1;
private javax.swing.JTextField jTextField1;
private javax.swing.JTextField jTextField2;
private javax.swing.JTextField jTextField3;
// End of variables declaration
}

```

OUTPUT:

10. Write a java program to display the following GUI.

CODE:

```
public class first_frame extends javax.swing.JFrame {
    private static final java.util.logging.Logger logger = java.util.logging.Logger.getLogger(first_frame.class.getName());

    public first_frame() {
        initComponents();
        jColorChooser1.setVisible(false);
    }
    @SuppressWarnings("unchecked")
    // <editor-fold defaultstate="collapsed" desc="Generated Code">
    private void initComponents() {
        jButton1 = new javax.swing.JButton();
        jButton2 = new javax.swing.JButton();
        jTextField1 = new javax.swing.JTextField();
        jColorChooser1 = new javax.swing.JColorChooser();
        setDefaultCloseOperation(javax.swing.WindowConstants.EXIT_ON_CLOSE);
        jButton1.setText("Set FontColor");
        jButton1.setBorder(new javax.swing.border.SoftBevelBorder(javax.swing.border.BevelBorder.RAISED, new java.awt.Color(0, 0, 255), new java.awt.Color(0, 0, 255), new java.awt.Color(0, 0, 255), new java.awt.Color(0, 0, 255)));
        jButton1.addActionListener(new java.awt.event.ActionListener() {
            public void actionPerformed(java.awt.event.ActionEvent evt) {
                jButton1ActionPerformed(evt);
            }
        });
        jButton2.setText("Set Background");
        jButton2.setBorder(new javax.swing.border.SoftBevelBorder(javax.swing.border.BevelBorder.RAISED, new java.awt.Color(0, 0, 255), new java.awt.Color(0, 0, 255), new java.awt.Color(0, 0, 255), new java.awt.Color(0, 0, 255)));
        jButton2.addActionListener(new java.awt.event.ActionListener() {
            public void actionPerformed(java.awt.event.ActionEvent evt) {
                jButton2ActionPerformed(evt);
            }
        });
        jTextField1.setFont(new java.awt.Font("Algerian", 1, 18)); // NOI18N
        jTextField1.setBorder(javax.swing.BorderFactory.createEtchedBorder(new java.awt.Color(0, 0, 255), null));
        javax.swing.GroupLayout layout = new javax.swing.GroupLayout(getContentPane());
        getContentPane().setLayout(layout);
        layout.setHorizontalGroup(
            layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)
                .addGroup(layout.createSequentialGroup()
                    .addGroup(javax.swing.GroupLayout.Alignment.TRAILING, layout.createSequentialGroup()
                        .addGap(73, 73, 73)
                        .addComponent(jButton1, javax.swing.GroupLayout.DEFAULT_SIZE, javax.swing.GroupLayout.DEFAULT_SIZE, Short.MAX_VALUE)
                    )
                )
        );
    }
}
```

```

        .addGap(28, 28, 28)
        .addComponent(jButton2, javax.swing.GroupLayout.DEFAULT_SIZE, javax.swing.GroupLayout.DEFAULT_SIZE,
Short.MAX_VALUE)
        .addGap(58, 58, 58))
        .addComponent(jColorChooser1, javax.swing.GroupLayout.Alignment.TRAILING, javax.swing.GroupLayout.DEFAULT_SIZE,
javax.swing.GroupLayout.DEFAULT_SIZE, Short.MAX_VALUE)
        .addGroup(layout.createSequentialGroup()
            .addContainerGap(javax.swing.GroupLayout.DEFAULT_SIZE, Short.MAX_VALUE)
            .addComponent(jTextField1)
            .addContainerGap(javax.swing.GroupLayout.DEFAULT_SIZE, Short.MAX_VALUE))
        );
    layout.setVerticalGroup(
        layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)
            .addGroup(javax.swing.GroupLayout.Alignment.TRAILING, layout.createSequentialGroup()
                .addContainerGap(javax.swing.GroupLayout.DEFAULT_SIZE, Short.MAX_VALUE)
                .addComponent(jTextField1, javax.swing.GroupLayout.DEFAULT_SIZE, 30, Short.MAX_VALUE)
                .addPreferredGap(javax.swing.LayoutStyle.ComponentPlacement.UNRELATED, javax.swing.GroupLayout.DEFAULT_SIZE,
Short.MAX_VALUE)
                .addComponent(jColorChooser1, javax.swing.GroupLayout.PREFERRED_SIZE, 265, Short.MAX_VALUE)
                .addGap(71, 71, 71)
                .addGroup(layout.createParallelGroup(javax.swing.GroupLayout.Alignment.BASELINE)
                    .addComponent(jButton1, javax.swing.GroupLayout.DEFAULT_SIZE, javax.swing.GroupLayout.DEFAULT_SIZE,
Short.MAX_VALUE)
                    .addComponent(jButton2, javax.swing.GroupLayout.DEFAULT_SIZE, javax.swing.GroupLayout.DEFAULT_SIZE,
Short.MAX_VALUE))
                .addGap(64, 64, 64))
            );
    pack();
}

private void jButton1ActionPerformed(java.awt.event.ActionEvent evt) {
    jColorChooser1.setVisible(true);
    jTextField1.setForeground(jColorChooser1.getColor());
}

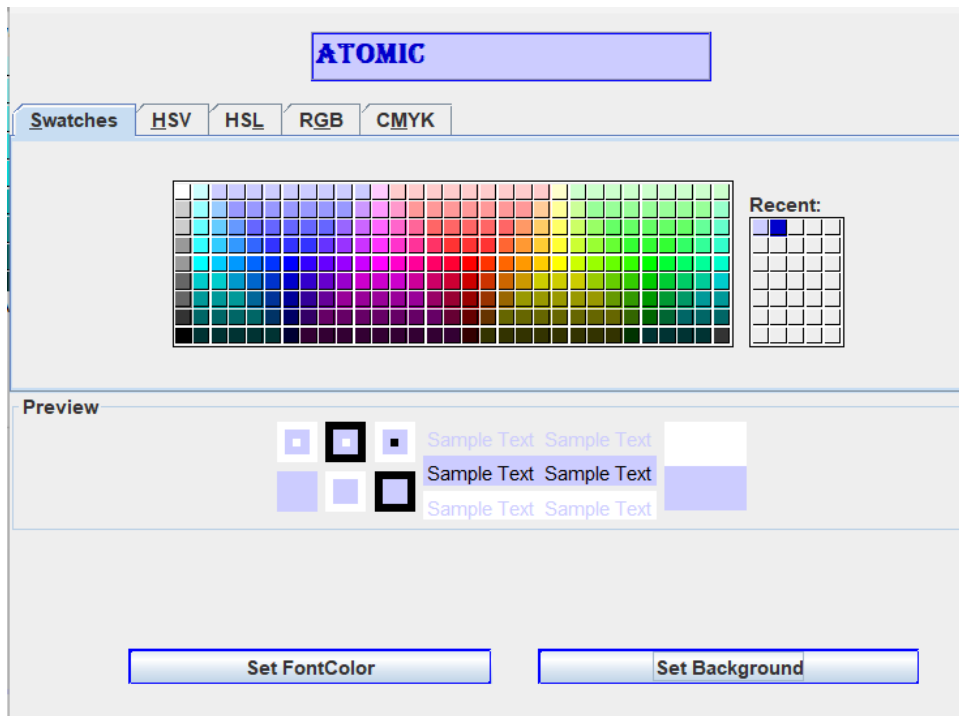
private void jButton2ActionPerformed(java.awt.event.ActionEvent evt) {
    jColorChooser1.setVisible(true);
    jTextField1.setBackground(jColorChooser1.getColor());
}

public static void main(String args[]) {
    java.awt.EventQueue.invokeLater(() -> new first_frame().setVisible(true));
}

// Variables declaration - do not modify
private javax.swing.JButton jButton1;
private javax.swing.JButton jButton2;
private javax.swing.JColorChooser jColorChooser1;
private javax.swing.JTextField jTextField1;
// End of variables declaration
}

```

OUTPUT:



11.Design a JFrame that initially appears with its background color equal to black. Each time the mouse is clicked, the background color becomes a little brighter. In other words, successive mouse clicks smoothly changes the background color from black to dark gray to white. (HW)

CODE:

```
package com.mycompany.swingprojects;
```

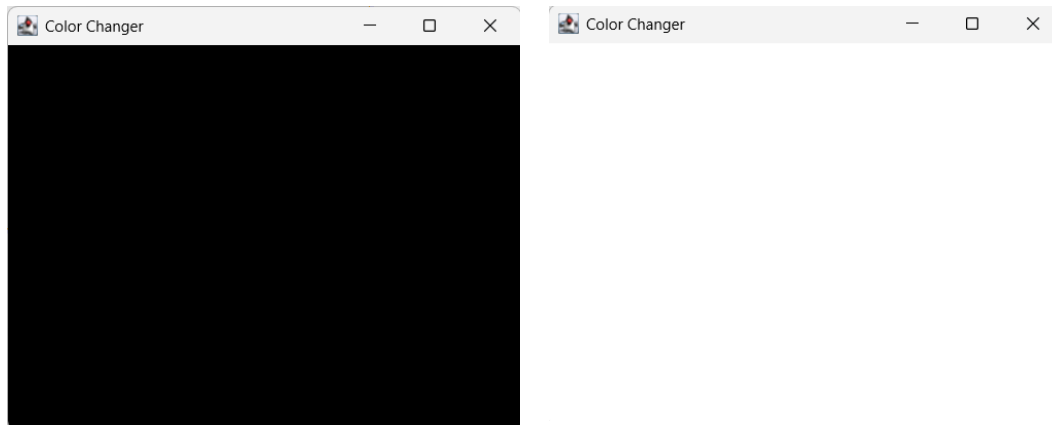
```
import java.awt.*;
```

```
public class color_changer_frame extends javax.swing.JFrame {
    private static final java.util.logging.Logger logger = java.util.logging.Logger.getLogger(color_changer_frame.class.getName());
    public color_changer_frame() {
        getContentPane().setBackground(Color.black);
        initComponents();
    }
    @SuppressWarnings("unchecked")
    // <editor-fold defaultstate="collapsed" desc="Generated Code">
    private void initComponents() {
        setDefaultCloseOperation(javax.swing.WindowConstants.EXIT_ON_CLOSE);
        setTitle("Color Changer");
        addMouseListener(new java.awt.event.MouseAdapter() {
            public void mouseClicked(java.awt.event.MouseEvent evt) {
                formMouseClicked(evt);
            }
        });
        javax.swing.GroupLayout layout = new javax.swing.GroupLayout(getContentPane());
        getContentPane().setLayout(layout);
        layout.setHorizontalGroup(
            layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)
                .addGroup(layout.createSequentialGroup()
                    .addGap(0, 400, Short.MAX_VALUE)
                    .addContainerGap())
        );
        layout.setVerticalGroup(
            layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)
                .addGroup(layout.createSequentialGroup()
                    .addGap(0, 300, Short.MAX_VALUE)
                    .addContainerGap())
        );
        pack();
    }
    private void formMouseClicked(java.awt.event.MouseEvent evt) {
```

```

        Color current = getContentPane().getBackground();
        getContentPane().setBackground(current.brighter());
    }
    public static void main(String args[]) {
        java.awt.EventQueue.invokeLater(() -> new color_changer_frame().setVisible(true));
    }
}

```

OUTPUT:

12. Write a java program to display the following GUI. Ask the user for quantity on placing order and show the final bill with all details in message box. (HW)

CODE:

```

package com.mycompany.swingprojects;

import javax.swing.JOptionPane;
import java.awt.Color;

public class pizza_order_frame extends javax.swing.JFrame {
    private static final java.util.logging.Logger logger = java.util.logging.Logger.getLogger(pizza_order_frame.class.getName());
    public pizza_order_frame() {
        getContentPane().setBackground(new Color(218,82,139));
        initComponents();
    }
    @SuppressWarnings("unchecked")
    // <editor-fold defaultstate="collapsed" desc="Generated Code">
    private void initComponents() {
        jLabel1 = new javax.swing.JLabel();
        jLabel2 = new javax.swing.JLabel();
        jLabel3 = new javax.swing.JLabel();
        jLabel4 = new javax.swing.JLabel();
        jLabel5 = new javax.swing.JLabel();
        jButton1 = new javax.swing.JButton();
        jScrollPane1 = new javax.swing.JScrollPane();
        jTextArea1 = new javax.swing.JTextArea();
        jTextField1 = new javax.swing.JTextField();
        jTextField2 = new javax.swing.JTextField();
        jComboBox1 = new javax.swing.JComboBox<>();
        jCheckBox1 = new javax.swing.JCheckBox();
        jCheckBox2 = new javax.swing.JCheckBox();
        jCheckBox3 = new javax.swing.JCheckBox();
        jCheckBox4 = new javax.swing.JCheckBox();
        setDefaultCloseOperation(javax.swing.WindowConstants.EXIT_ON_CLOSE);
        setBackground(new java.awt.Color(218, 82, 139));
        jLabel1.setFont(new java.awt.Font("STLiti", 1, 14)); // NOI18N
        jLabel1.setForeground(new java.awt.Color(0, 171, 231));
        jLabel1.setText("Name:");
        jLabel2.setFont(new java.awt.Font("STLiti", 1, 14)); // NOI18N
        jLabel2.setForeground(new java.awt.Color(0, 171, 231));
        jLabel2.setText("Address:");
        jLabel3.setFont(new java.awt.Font("STLiti", 1, 14)); // NOI18N

```



```

jLabel3.setForeground(new java.awt.Color(0, 171, 231));
jLabel3.setText("Phone No:");
jLabel4.setFont(new java.awt.Font("STLiti", 1, 14)); // NOI18N
jLabel4.setForeground(new java.awt.Color(0, 171, 231));
jLabel4.setText("Crust:");
jLabel5.setFont(new java.awt.Font("STLiti", 1, 14)); // NOI18N
jLabel5.setForeground(new java.awt.Color(0, 171, 231));
jLabel5.setText("Toppings:");
jButton1.setBackground(new java.awt.Color(204, 204, 255));
jButton1.setFont(jLabel1.getFont());
jButton1.setForeground(new java.awt.Color(102, 204, 255));
jButton1.setText("Order");
jButton1.addActionListener(new java.awt.event.ActionListener() {
    public void actionPerformed(java.awt.event.ActionEvent evt) {
        jButton1ActionPerformed(evt);
    }
});
jTextArea1.setColumns(20);
jTextArea1.setRows(5);
jScrollPane1.setViewportView(jTextArea1);
jComboBox1.setModel(new javax.swing.DefaultComboBoxModel<>(new String[] { "Choose option...", "Hand Tossed", "Thin
Crust", "Thick Crust", "Pan Pizza" }));
jCheckBox1.setBackground(new java.awt.Color(218, 82, 139));
jCheckBox1.setText("Pepperoni");
jCheckBox2.setBackground(new java.awt.Color(218, 82, 139));
jCheckBox2.setText("Mushrooms");
jCheckBox3.setBackground(new java.awt.Color(218, 82, 139));
jCheckBox3.setText("Sausage");
jCheckBox4.setBackground(new java.awt.Color(218, 82, 139));
jCheckBox4.setText("Bacon");
javax.swing.GroupLayout layout = new javax.swing.GroupLayout(getContentPane());
getContentPane().setLayout(layout);
layout.setHorizontalGroup(
    layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)
        .addGroup(layout.createSequentialGroup()
            .addGap(35, 35, 35)
            .addGroup(layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)
                .addComponent(jButton1)
                .addGap(0, 0, Short.MAX_VALUE))
            .addGap(18, 18, 18)
            .addGroup(layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)
                .addComponent(jScrollPane1)
                .addComponent(jTextField1)
                .addComponent(jTextField2)
                .addComponent(jComboBox1, 0, Short.MAX_VALUE))
            .addGap(0, 0, Short.MAX_VALUE))
        .addGroup(layout.createParallelGroup(javax.swing.GroupLayout.Alignment.TRAILING, false)
            .addComponent(jCheckBox4, javax.swing.GroupLayout.Alignment.LEADING,
                javax.swing.GroupLayout.DEFAULT_SIZE, javax.swing.GroupLayout.DEFAULT_SIZE, Short.MAX_VALUE)
            .addComponent(jCheckBox3, javax.swing.GroupLayout.Alignment.LEADING,
                javax.swing.GroupLayout.DEFAULT_SIZE, javax.swing.GroupLayout.DEFAULT_SIZE, Short.MAX_VALUE)
            .addComponent(jCheckBox2, javax.swing.GroupLayout.Alignment.LEADING,
                javax.swing.GroupLayout.DEFAULT_SIZE, javax.swing.GroupLayout.DEFAULT_SIZE, Short.MAX_VALUE)
            .addGroup(layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)
                .addComponent(jLabel1, javax.swing.GroupLayout.DEFAULT_SIZE, javax.swing.GroupLayout.DEFAULT_SIZE, Short.MAX_VALUE)
                .addComponent(jLabel2, javax.swing.GroupLayout.DEFAULT_SIZE, javax.swing.GroupLayout.DEFAULT_SIZE, Short.MAX_VALUE)
                .addComponent(jLabel3, javax.swing.GroupLayout.DEFAULT_SIZE, javax.swing.GroupLayout.DEFAULT_SIZE, Short.MAX_VALUE)
                .addComponent(jLabel4, javax.swing.GroupLayout.DEFAULT_SIZE, javax.swing.GroupLayout.DEFAULT_SIZE, Short.MAX_VALUE)
                .addComponent(jLabel5, javax.swing.GroupLayout.DEFAULT_SIZE, javax.swing.GroupLayout.DEFAULT_SIZE, Short.MAX_VALUE))
            .addGap(0, 0, Short.MAX_VALUE))
    )

```

```

        .addComponent(jCheckBox1, javax.swing.GroupLayout.Alignment.LEADING,
javax.swing.GroupLayout.DEFAULT_SIZE, javax.swing.GroupLayout.DEFAULT_SIZE, Short.MAX_VALUE)))
        .addGap(221, 221, 221)))
    );
    layout.setVerticalGroup(
        layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)
        .addGroup(layout.createSequentialGroup()
            .addGap(48, 48, 48)
            .addGroup(layout.createParallelGroup(javax.swing.GroupLayout.Alignment.BASELINE)
                .addComponent(jLabel1)
                .addComponent(jTextField1, javax.swing.GroupLayout.PREFERRED_SIZE, javax.swing.GroupLayout.DEFAULT_SIZE,
javax.swing.GroupLayout.PREFERRED_SIZE))
            .addPreferredGap(javax.swing.LayoutStyle.ComponentPlacement.UNRELATED)
            .addGroup(layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)
                .addComponent(jScrollPane1, javax.swing.GroupLayout.PREFERRED_SIZE, 46,
javax.swing.GroupLayout.PREFERRED_SIZE)
                .addComponent(jLabel2))
            .addPreferredGap(javax.swing.LayoutStyle.ComponentPlacement.RELATED, 16, Short.MAX_VALUE)
            .addGroup(layout.createParallelGroup(javax.swing.GroupLayout.Alignment.BASELINE)
                .addComponent(jLabel3)
                .addComponent(jTextField2, javax.swing.GroupLayout.PREFERRED_SIZE, javax.swing.GroupLayout.DEFAULT_SIZE,
javax.swing.GroupLayout.PREFERRED_SIZE))
            .addGap(16, 16, 16)
            .addGroup(layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)
                .addComponent(jLabel4)
                .addComponent(jComboBox1, javax.swing.GroupLayout.PREFERRED_SIZE, javax.swing.GroupLayout.DEFAULT_SIZE,
javax.swing.GroupLayout.PREFERRED_SIZE))
            .addGap(27, 27, 27)
            .addGroup(layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)
                .addComponent(jLabel5)
                .addComponent(jCheckBox1))
            .addPreferredGap(javax.swing.LayoutStyle.ComponentPlacement.RELATED)
            .addComponent(jCheckBox2)
            .addPreferredGap(javax.swing.LayoutStyle.ComponentPlacement.RELATED)
            .addComponent(jCheckBox3)
            .addPreferredGap(javax.swing.LayoutStyle.ComponentPlacement.RELATED)
            .addComponent(jCheckBox4)
            .addGap(26, 26, 26)
            .addComponent(jButton1)
            .addGap(146, 146, 146))
    );
    pack();
}

```

```

private void jButton1ActionPerformed(java.awt.event.ActionEvent evt) {
    String name = jTextField1.getText();
    String add = jTextArea1.getText();
    String pno = jTextField2.getText();
    String sclass = (String) jComboBox1.getSelectedItem();
    String toppings = "Toppings";
    if(jCheckBox1.isSelected()) toppings+="Pepperoni";
    if(jCheckBox2.isSelected()) toppings+="Mushrooms";
    if(jCheckBox3.isSelected()) toppings+="Sausage";
    if(jCheckBox4.isSelected()) toppings+="Bacon";
    String order = "Name: " + name +
        "\nPhone No: " + pno +
        "\nAddress: " + add +
        "\nCrust: " + sclass +
        "\nToppings: " + toppings;

```

```

    JOptionPane.showMessageDialog(this, order, "Order Placed",
    JOptionPane.INFORMATION_MESSAGE);
}

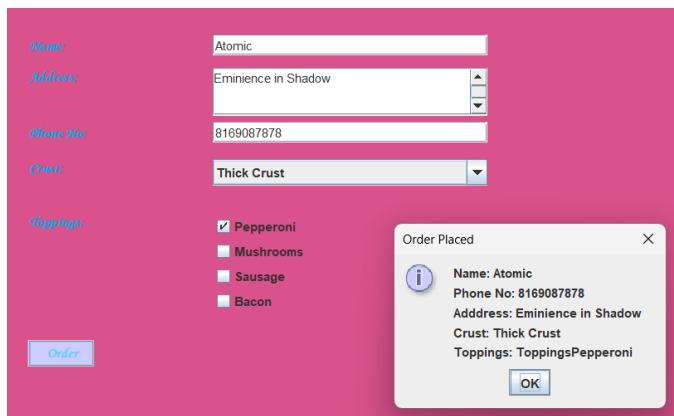
```

```

public static void main(String args[]) {
    java.awt.EventQueue.invokeLater(() -> new pizza_order_frame().setVisible(true));
}
// Variables declaration - do not modify
private javax.swing.JButton jButton1;
private javax.swing.JCheckBox jCheckBox1;
private javax.swing.JCheckBox jCheckBox2;
private javax.swing.JCheckBox jCheckBox3;
private javax.swing.JCheckBox jCheckBox4;
private javax.swing.JComboBox<String> jComboBox1;
private javax.swing.JLabel jLabel1;
private javax.swing.JLabel jLabel2;
private javax.swing.JLabel jLabel3;
private javax.swing.JLabel jLabel4;
private javax.swing.JLabel jLabel5;
private javax.swing.JScrollPane jScrollPane1;
private javax.swing.JTextArea jTextArea1;
private javax.swing.JTextField jTextField1;
private javax.swing.JTextField jTextField2;
// End of variables declaration
}

```

OUTPUT:



13. Write a java program to display the content of the text file using the following GUI. (hint: use JFileChooser)

CODE:

```

package com.mycompany.swingprojects;
import java.io.*;
import javax.swing.JFileChooser;
public class third_frame extends javax.swing.JFrame {
    private static final java.util.logging.Logger logger = java.util.logging.Logger.getLogger(third_frame.class.getName());
    public third_frame() {
        initComponents();
    }
    @SuppressWarnings("unchecked")
    // <editor-fold defaultstate="collapsed" desc="Generated Code">
    private void initComponents() {
        jButton1 = new javax.swing.JButton();
        jScrollPane1 = new javax.swing.JScrollPane();
        jTextArea1 = new javax.swing.JTextArea();
        jLabel1 = new javax.swing.JLabel();
        setDefaultCloseOperation(javax.swing.WindowConstants.EXIT_ON_CLOSE);
        setTitle("File Copying Tool");
        setBackground(new java.awt.Color(102, 102, 102));
        jButton1.setText("OPEN");
        jButton1.addActionListener(new java.awt.event.ActionListener() {

```

```

        public void actionPerformed(java.awt.event.ActionEvent evt) {
            jButton1ActionPerformed(evt);
        }
    });
    jTextArea1.setBackground(new java.awt.Color(0, 204, 51));
    jTextArea1.setColumns(20);
    jTextArea1.setFont(new java.awt.Font("Calisto MT", 1, 14)); // NOI18N
    jTextArea1.setForeground(new java.awt.Color(255, 0, 0));
    jTextArea1.setRows(5);
    jScrollPane1.setViewportView(jTextArea1);
    jLabel1.setBackground(new java.awt.Color(0, 255, 0));
    jLabel1.setFont(new java.awt.Font("Segoe Print", 1, 14)); // NOI18N
    jLabel1.setText("Text Copy Editor");
    javax.swing.GroupLayout layout = new javax.swing.GroupLayout(getContentPane());
    getContentPane().setLayout(layout);
    layout.setHorizontalGroup(
        layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)
            .addGroup(layout.createSequentialGroup()
                .add(layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)
                    .addGroup(layout.createSequentialGroup()
                        .add(layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)
                            .add(layout.createSequentialGroup()
                                .addGap(173, 173, 173)
                                .addComponent(jButton1))
                            .add(layout.createSequentialGroup()
                                .addGap(80, 80, 80)
                                .addComponent(jScrollPane1, javax.swing.GroupLayout.DEFAULT_SIZE, 275, Short.MAX_VALUE)))
                        .addGap(88, 88, 88))
                    .addGroup(layout.createSequentialGroup()
                        .addGap(148, 148, 148)
                        .addComponent(jLabel1)
                        .addContainerGap(javax.swing.GroupLayout.DEFAULT_SIZE, Short.MAX_VALUE))
                    .add(layout.createSequentialGroup()
                        .addGap(60, 60, 60)
                        .addComponent(jLabel1, javax.swing.GroupLayout.DEFAULT_SIZE, javax.swing.GroupLayout.DEFAULT_SIZE, Short.MAX_VALUE)
                        .addGap(57, 57, 57)
                        .addComponent(jScrollPane1, javax.swing.GroupLayout.PREFERRED_SIZE, javax.swing.GroupLayout.DEFAULT_SIZE, javax.swing.GroupLayout.PREFERRED_SIZE)
                        .addGap(29, 29, 29)
                        .addComponent(jButton1)
                        .addGap(83, 83, 83))
                .addContainerGap())
    );
    pack();
}

```

```

private void jButton1ActionPerformed(java.awt.event.ActionEvent evt) {

```

```

    JFileChooser fc = new JFileChooser();

```

```

    int r = fc.showOpenDialog(this);

```

```

    if(r==JFileChooser.APPROVE_OPTION){

```

```

        int i;

```

```

        String p = fc.getSelectedFile().getPath();

```

```

        String msg = "";

```

```

        try{

```

```

            File f = new File(p);

```

```

            FileReader fr = new FileReader(f);

```

```

            while((i = fr.read()) != -1){

```

```

                msg = msg + (char)i;

```

```

            }

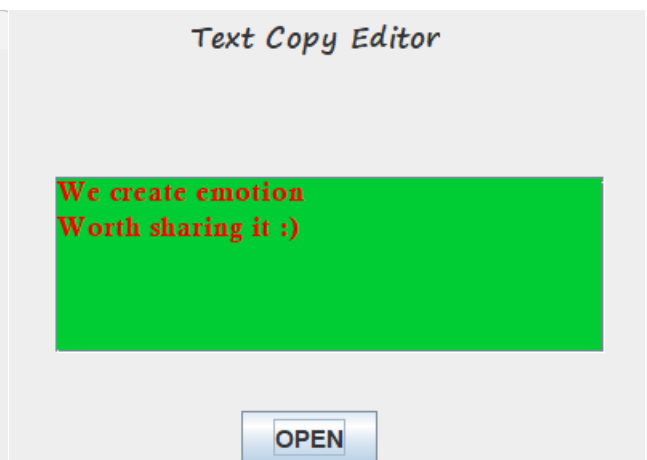
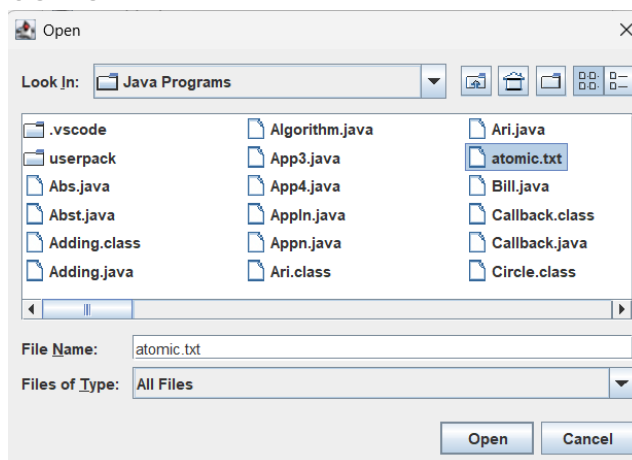
```

```

            jTextArea1.setText(msg);

```

```
    }  
    catch(Exception e){  
        System.out.println(e);  
    }  
}  
}  
public static void main(String args[]) {  
    java.awt.EventQueue.invokeLater(() -> new third_frame().setVisible(true));  
}  
// Variables declaration - do not modify  
private javax.swing.JButton jButton1;  
private javax.swing.JLabel jLabel1;  
private javax.swing.JScrollPane jScrollPane1;  
private javax.swing.JTextArea jTextArea1;  
// End of variables declaration  
}
```

OUTPUT:

14. Write a java program to display the following GUI.**CODE:**

```

package com.mycompany.swingprojects;
public class multiple_jframe extends javax.swing.JFrame {
    public multiple_jframe() {
        initComponents();
    }
    @SuppressWarnings("unchecked")
    // <editor-fold defaultstate="collapsed" desc="Generated Code">
    private void initComponents() {
        jMenuBar1 = new javax.swing.JMenuBar();
        jMenu1 = new javax.swing.JMenu();
        jMenu2 = new javax.swing.JMenu();
        jMenu3 = new javax.swing.JMenu();
        jMenuItem1 = new javax.swing.JMenuItem();
        jMenuItem2 = new javax.swing.JMenuItem();
        jMenuItem3 = new javax.swing.JMenuItem();
        setDefaultCloseOperation(javax.swing.WindowConstants.EXIT_ON_CLOSE);
        setTitle("Dream Window");
        jMenu1.setText("File");
        jMenuBar1.add(jMenu1);
        jMenu2.setText("Edit");
        jMenuBar1.add(jMenu2);
        jMenu3.setText("Search");
        jMenuItem1.setText("Colors");
        jMenuItem1.addActionListener(new java.awt.event.ActionListener() {
            public void actionPerformed(java.awt.event.ActionEvent evt) {
                jMenuItem1ActionPerformed(evt);
            }
        });
        jMenuItem3.add(jMenuItem1);
        jMenuItem2.setText("Cities");
        jMenuItem2.addActionListener(new java.awt.event.ActionListener() {
            public void actionPerformed(java.awt.event.ActionEvent evt) {
                jMenuItem2ActionPerformed(evt);
            }
        });
        jMenuItem3.add(jMenuItem2);

        jMenuItem3.setText("Cars");
        jMenuItem3.addActionListener(new java.awt.event.ActionListener() {
            public void actionPerformed(java.awt.event.ActionEvent evt) {
                jMenuItem3ActionPerformed(evt);
            }
        });
        jMenuItem3.add(jMenuItem3);
        jMenuBar1.add(jMenu3);
        setJMenuBar(jMenuBar1);
        javax.swing.GroupLayout layout = new javax.swing.GroupLayout(getContentPane());
        getContentPane().setLayout(layout);
        layout.setHorizontalGroup(
            layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)
                .addGroup(layout.createSequentialGroup()
                    .addGap(0, 400, Short.MAX_VALUE)
                )
        );
        layout.setVerticalGroup(
            layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)
                .addGroup(layout.createSequentialGroup()
                    .addGap(0, 277, Short.MAX_VALUE)
                )
        );
        pack();
    }

    private void jMenuItem2ActionPerformed(java.awt.event.ActionEvent evt) {
        city_frame c = new city_frame();
        c.setVisible(true);
    }
}

```

```

private void jMenuItem1ActionPerformed(java.awt.event.ActionEvent evt) {
    color_frame cf = new color_frame();
    cf.setVisible(true);
}
private void jMenuItem3ActionPerformed(java.awt.event.ActionEvent evt) {
    car_frame cr = new car_frame();
    cr.setVisible(true);
}
public static void main(String args[]) {
    java.awt.EventQueue.invokeLater(() -> new multiple_jframe().setVisible(true));
}
// Variables declaration - do not modify
private javax.swing.JMenu jMenuItem1;
private javax.swing.JMenu jMenuItem2;
private javax.swing.JMenu jMenuItem3;
private javax.swing.JMenuBar jMenuItemBar1;
private javax.swing.JMenuItem jMenuItem1;
private javax.swing.JMenuItem jMenuItem2;
private javax.swing.JMenuItem jMenuItem3;
// End of variables declaration
}

```

Color Frame CODE:

```

Citypackage com.mycompany.swingprojects;
import java.awt.*;
public class color_frame extends javax.swing.JFrame {
    public color_frame() {
        getContentPane().setBackground(Color.black);
        initComponents();
    }
    private void formMouseClicked(java.awt.event.MouseEvent evt) {
        Color current = getContentPane().getBackground();
        getContentPane().setBackground(current.brighter());
    }
    public static void main(String args[]) {
        java.awt.EventQueue.invokeLater(() -> new color_frame().setVisible(true));
    }
    private javax.swing.JLabel jLabel1;
}

```

Frame CODE:

```

package com.mycompany.swingprojects;
import javax.swing.JOptionPane;
public class city_frame extends javax.swing.JFrame {
    public city_frame() {
        initComponents();
    }
    private void jButton1ActionPerformed(java.awt.event.ActionEvent evt) {
        String msg = jTextField1.getText();
        JOptionPane.showMessageDialog(this, "Keep working hard to achieve your dreams!", msg,
JOptionPane.INFORMATION_MESSAGE);
    }
    public static void main(String args[]) {
        java.awt.EventQueue.invokeLater(() -> new city_frame().setVisible(true));
    }
    private javax.swing.JButton jButton1;
    private javax.swing.JLabel jLabel1;
    private javax.swing.JTextField jTextField1;
}

```

Car Frame CODE:

```

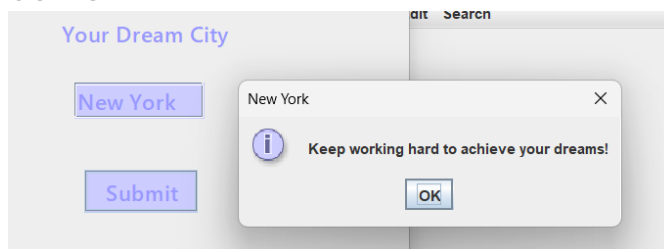
package com.mycompany.swingprojects;
import javax.swing.*;
import java.awt.*;

```

```

public class car_frame extends javax.swing.JFrame {
    public car_frame() {
        getContentPane().setBackground(Color.white);
        initComponents();
    }
    private void jButton4ActionPerformed(java.awt.event.ActionEvent evt) {
        JOptionPane.showMessageDialog(null, "You are wild!");
    }
    private void jButton1ActionPerformed(java.awt.event.ActionEvent evt) {
        JOptionPane.showMessageDialog(null, "Speed is the only thing!");
    }
    private void jButton2ActionPerformed(java.awt.event.ActionEvent evt) {
        JOptionPane.showMessageDialog(null, "Flexing!!");
    }
    private void jButton3ActionPerformed(java.awt.event.ActionEvent evt) {
        JOptionPane.showMessageDialog(null, "More Acceleration!");
    }
    public static void main(String args[]) {
        java.awt.EventQueue.invokeLater(() -> new car_frame().setVisible(true));
    }
    private javax.swing.JLabel jLabel1;
    private javax.swing.JRadioButton jRadioButton1;
    private javax.swing.JRadioButton jRadioButton2;
    private javax.swing.JRadioButton jRadioButton3;
    private javax.swing.JRadioButton jRadioButton4;
}

```

OUTPUT:

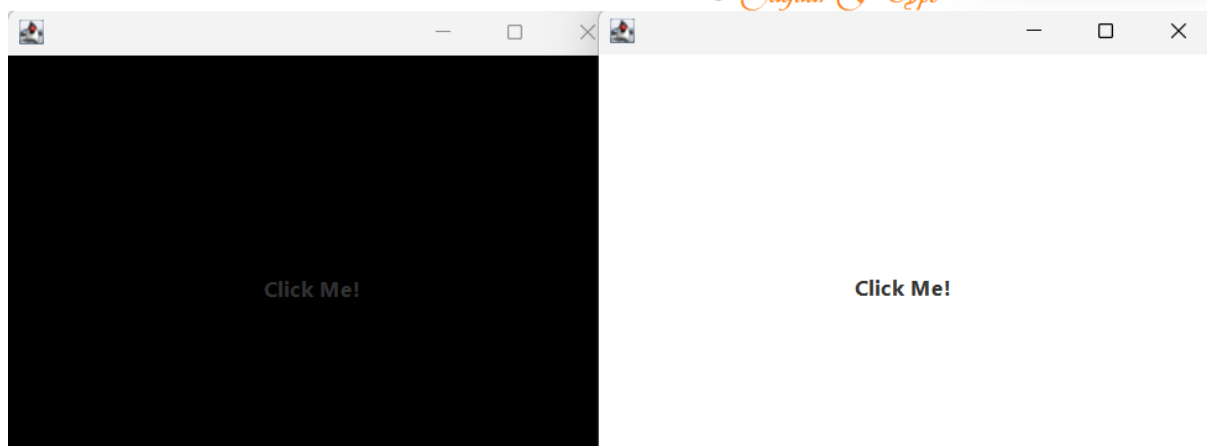
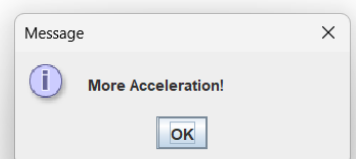
Choose Your Dream car!

☐ *Lamborghini Huracan EVO*

☐ *Ferrari 488*

☒ *McLaren 750S*

☐ *Jaguar F-Type*




```
jLabel1.setFont(new java.awt.Font("sansserif", 1, 18)); // NOI18N
jLabel1.setForeground(new java.awt.Color(204, 204, 204));
jLabel1.setHorizontalAlignment(javax.swing.SwingConstants.CENTER);
jLabel1.setText("Voting Machine");
jLabel3.setBackground(new java.awt.Color(40, 40, 40));
jLabel3.setFont(new java.awt.Font("Segoe UI", 1, 14)); // NOI18N
jLabel3.setForeground(new java.awt.Color(102, 255, 0));
jLabel3.setHorizontalAlignment(javax.swing.SwingConstants.CENTER);
jLabel4.setBackground(new java.awt.Color(40, 40, 40));
jLabel4.setFont(new java.awt.Font("Segoe UI", 1, 14)); // NOI18N
jLabel4.setForeground(new java.awt.Color(102, 255, 0));
jLabel4.setHorizontalAlignment(javax.swing.SwingConstants.CENTER);
jLabel5.setBackground(new java.awt.Color(40, 40, 40));
jLabel5.setFont(new java.awt.Font("Segoe UI", 1, 14)); // NOI18N
jLabel5.setForeground(new java.awt.Color(102, 255, 0));
jLabel5.setHorizontalAlignment(javax.swing.SwingConstants.CENTER);
jLabel6.setBackground(new java.awt.Color(40, 40, 40));
jLabel6.setFont(new java.awt.Font("Segoe UI", 1, 14)); // NOI18N
jLabel6.setForeground(new java.awt.Color(102, 255, 0));
jLabel6.setHorizontalAlignment(javax.swing.SwingConstants.CENTER);
jRadioButton1.setBackground(new java.awt.Color(40, 40, 40));
buttonGroup1.add(jRadioButton1);
jRadioButton1.setFont(new java.awt.Font("Segoe UI", 1, 14)); // NOI18N
jRadioButton1.setForeground(new java.awt.Color(102, 255, 0));
jRadioButton1.setText("BJP");
jRadioButton1.addActionListener(new java.awt.event.ActionListener() {
    public void actionPerformed(java.awt.event.ActionEvent evt) {
        jRadioButton1ActionPerformed(evt);
    }
});
jRadioButton2.setBackground(new java.awt.Color(40, 40, 40));
buttonGroup1.add(jRadioButton2);
jRadioButton2.setFont(new java.awt.Font("Segoe UI", 1, 14)); // NOI18N
jRadioButton2.setForeground(new java.awt.Color(102, 255, 0));
jRadioButton2.setText("Shiv Sena");
jRadioButton3.setBackground(new java.awt.Color(40, 40, 40));
buttonGroup1.add(jRadioButton3);
jRadioButton3.setFont(new java.awt.Font("Segoe UI", 1, 14)); // NOI18N
jRadioButton3.setForeground(new java.awt.Color(102, 255, 0));
jRadioButton3.setText("Kanoha");
jRadioButton4.setBackground(new java.awt.Color(40, 40, 40));
buttonGroup1.add(jRadioButton4);
jRadioButton4.setFont(new java.awt.Font("Segoe UI", 1, 14)); // NOI18N
jRadioButton4.setForeground(new java.awt.Color(102, 255, 0));
jRadioButton4.setText("Sand village");
jRadioButton4.addActionListener(new java.awt.event.ActionListener() {
    public void actionPerformed(java.awt.event.ActionEvent evt) {
        jRadioButton4ActionPerformed(evt);
    }
});
jRadioButton5.setBackground(new java.awt.Color(40, 40, 40));
buttonGroup1.add(jRadioButton5);
jRadioButton5.setFont(new java.awt.Font("Segoe UI", 1, 14)); // NOI18N
jRadioButton5.setForeground(new java.awt.Color(102, 255, 0));
jRadioButton5.setText("Uzumaki");
jRadioButton5.addActionListener(new java.awt.event.ActionListener() {
    public void actionPerformed(java.awt.event.ActionEvent evt) {
        jRadioButton5ActionPerformed(evt);
    }
});
jLabel7.setBackground(new java.awt.Color(40, 40, 40));
jLabel7.setFont(new java.awt.Font("Segoe UI", 1, 14)); // NOI18N
jLabel7.setForeground(new java.awt.Color(102, 255, 0));
jLabel7.setHorizontalAlignment(javax.swing.SwingConstants.CENTER);
jButton1.setBackground(new java.awt.Color(40, 40, 40));
jButton1.setFont(new java.awt.Font("Segoe UI", 1, 14)); // NOI18N
```

```

jButton1.setForeground(new java.awt.Color(102, 255, 0));
jButton1.setText("Submit");
jButton1.addActionListener(new java.awt.event.ActionListener() {
    public void actionPerformed(java.awt.event.ActionEvent evt) {
        jButton1ActionPerformed(evt);
    }
});

jButton2.setBackground(new java.awt.Color(40, 40, 40));
jButton2.setFont(new java.awt.Font("Segoe UI", 1, 14)); // NOI18N
jButton2.setForeground(new java.awt.Color(102, 255, 0));
jButton2.setText("Statistics");
jButton2.addActionListener(new java.awt.event.ActionListener() {
    public void actionPerformed(java.awt.event.ActionEvent evt) {
        jButton2ActionPerformed(evt);
    }
});

javax.swing.GroupLayout layout = new javax.swing.GroupLayout(getContentPane());
getContentPane().setLayout(layout);
layout.setHorizontalGroup(
    layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)
        .addGroup(layout.createSequentialGroup()
            .addGap(104, 104, 104)
            .addGroup(layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)
                .addComponent(jLabel6, javax.swing.GroupLayout.DEFAULT_SIZE, javax.swing.GroupLayout.DEFAULT_SIZE, Short.MAX_VALUE)
                .addComponent(jLabel4, javax.swing.GroupLayout.DEFAULT_SIZE, javax.swing.GroupLayout.DEFAULT_SIZE, Short.MAX_VALUE)
                .addComponent(jLabel5, javax.swing.GroupLayout.DEFAULT_SIZE, javax.swing.GroupLayout.DEFAULT_SIZE, Short.MAX_VALUE)
                .addComponent(jLabel7, javax.swing.GroupLayout.DEFAULT_SIZE, 94, Short.MAX_VALUE)
                .addComponent(jLabel3, javax.swing.GroupLayout.DEFAULT_SIZE, javax.swing.GroupLayout.DEFAULT_SIZE, Short.MAX_VALUE))
            .addGap(104, 104, 104)
            .addGroup(layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)
                .addComponent(jRadioButton2, javax.swing.GroupLayout.PREFERRED_SIZE, 98, javax.swing.GroupLayout.PREFERRED_SIZE)
                .addComponent(jRadioButton1, javax.swing.GroupLayout.PREFERRED_SIZE, 98, javax.swing.GroupLayout.PREFERRED_SIZE)
                .addComponent(jRadioButton3, javax.swing.GroupLayout.PREFERRED_SIZE, 98, javax.swing.GroupLayout.PREFERRED_SIZE)
                .addComponent(jRadioButton5, javax.swing.GroupLayout.PREFERRED_SIZE, 98, javax.swing.GroupLayout.PREFERRED_SIZE)
                .addComponent(jRadioButton4, javax.swing.GroupLayout.PREFERRED_SIZE, 114, javax.swing.GroupLayout.PREFERRED_SIZE))
            .addGap(144, 144, 144)
            .addComponent(jLabel1, javax.swing.GroupLayout.PREFERRED_SIZE, 200, javax.swing.GroupLayout.PREFERRED_SIZE)
            .addGap(146, 146, 146)
            .addComponent(jButton1)
            .addGap(49, 49, 49)
            .addComponent(jButton2))
        .addContainerGap(89, Short.MAX_VALUE))
);
layout.setVerticalGroup(
    layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)
        .addGroup(layout.createSequentialGroup()
            .addGap(25, 25, 25)
            .addGroup(layout.createParallelGroup(javax.swing.GroupLayout.Alignment.BASELINE)
                .addComponent(jRadioButton1)

```

```

        .addComponent(jLabel3, javax.swing.GroupLayout.PREFERRED_SIZE, 28, javax.swing.GroupLayout.PREFERRED_SIZE))
        .addGap(18, 18, 18)
        .addGroup(layout.createParallelGroup(javax.swing.GroupLayout.Alignment.BASELINE)
            .addComponent(jLabel4, javax.swing.GroupLayout.PREFERRED_SIZE, 28, javax.swing.GroupLayout.PREFERRED_SIZE)
            .addComponent(jRadioButton2))
        .addGap(21, 21, 21)
        .addGroup(layout.createParallelGroup(javax.swing.GroupLayout.Alignment.BASELINE)
            .addComponent(jRadioButton3)
            .addComponent(jLabel5, javax.swing.GroupLayout.PREFERRED_SIZE, 28, javax.swing.GroupLayout.PREFERRED_SIZE))
        .addGroup(layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)
            .addGroup(layout.createSequentialGroup()
                .addGap(72, 72, 72)
                .addComponent(jRadioButton5))
            .addGroup(layout.createSequentialGroup()
                .addGap(23, 23, 23)
                .addGroup(layout.createParallelGroup(javax.swing.GroupLayout.Alignment.BASELINE)
                    .addComponent(jLabel6, javax.swing.GroupLayout.PREFERRED_SIZE, 28,
javax.swing.GroupLayout.PREFERRED_SIZE)
                    .addComponent(jRadioButton4))
                .addGap(18, 18, 18)
                .addComponent(jLabel7, javax.swing.GroupLayout.PREFERRED_SIZE, 28, javax.swing.GroupLayout.PREFERRED_SIZE)))
            .addGap(53, 53, 53)
            .addGroup(layout.createParallelGroup(javax.swing.GroupLayout.Alignment.BASELINE)
                .addComponent(jButton1)
                .addComponent(jButton2))
            .addContainerGap(23, Short.MAX_VALUE))
        );
    pack();
}

private void jButton1ActionPerformed(java.awt.event.ActionEvent evt) {
    if (jRadioButton1.isSelected()) {
        voteCounts[0]++;
        JOptionPane.showMessageDialog(this, "You voted for BJP");
    } else if (jRadioButton2.isSelected()) {
        voteCounts[1]++;
        JOptionPane.showMessageDialog(this, "You voted for Shiv Sena.");
    } else if (jRadioButton3.isSelected()) {
        voteCounts[2]++;
        JOptionPane.showMessageDialog(this, "You voted for Kanoha.");
    } else if (jRadioButton4.isSelected()) {
        voteCounts[3]++;
        JOptionPane.showMessageDialog(this, "You voted for Sand Village.");
    } else if (jRadioButton5.isSelected()) {
        voteCounts[4]++;
        JOptionPane.showMessageDialog(this, "You voted for Uzumaki.");
    } else {
        JOptionPane.showMessageDialog(this, "Please select a candidate.");
    }
    buttonGroup1.clearSelection();
}

private void jButton2ActionPerformed(java.awt.event.ActionEvent evt) {
    // TODO add your handling code here:
    String message = "Vote Statistics:\n"
        + "BJP: " + voteCounts[0] + "\n"
        + "Shiv Sena: " + voteCounts[1] + "\n"
        + "Kanoha: " + voteCounts[2] + "\n"

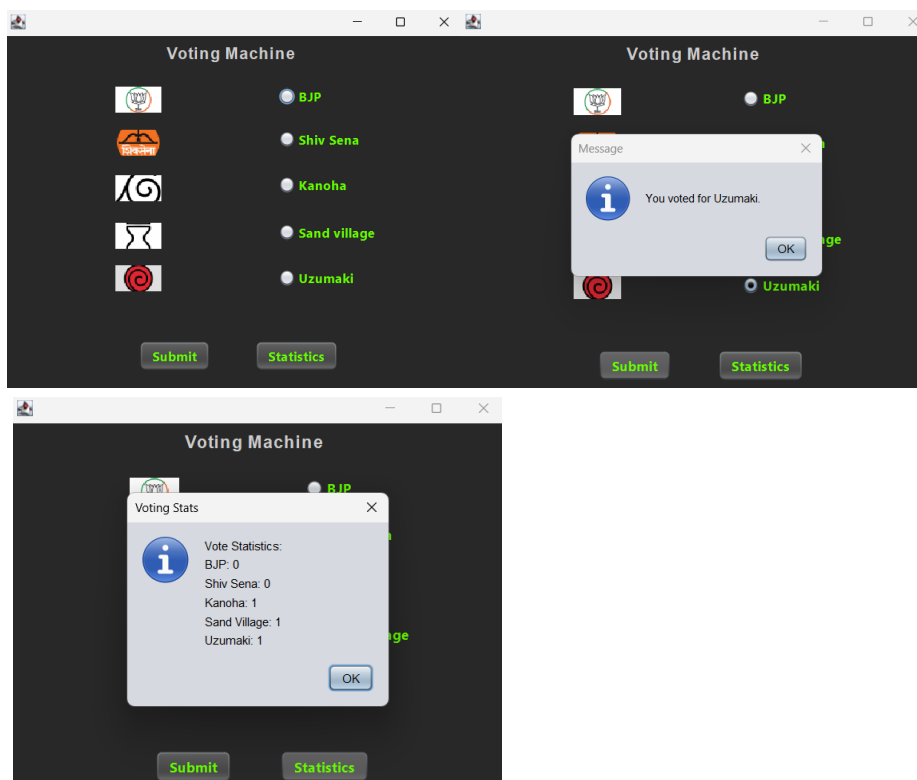
```

```

        + "Sand Village: " + voteCounts[3] + "\n"
        + "Uzumaki: " + voteCounts[4];
    JOptionPane.showMessageDialog(this, message, "Voting Stats",
JOptionPane.INFORMATION_MESSAGE);
}
public static void main(String args[]) {
    try {
        for (javax.swing.UIManager.LookAndFeelInfo info : javax.swing.UIManager.getInstalledLookAndFeels()) {
            if ("Nimbus".equals(info.getName())) {
                javax.swing.UIManager.setLookAndFeel(info.getClassName());
                break;
            }
        }
    } catch (ReflectiveOperationException | javax.swing.UnsupportedLookAndFeelException ex) {
        logger.log(java.util.logging.Level.SEVERE, null, ex);
    }
    java.awt.EventQueue.invokeLater(() -> new Voting_machine().setVisible(true));
}
// Variables declaration - do not modify
private javax.swing.ButtonGroup buttonGroup1;
private javax.swing.JButton jButton1;
private javax.swing.JButton jButton2;
private javax.swing.JLabel jLabel1;
private javax.swing.JLabel jLabel3;
private javax.swing.JLabel jLabel4;
private javax.swing.JLabel jLabel5;
private javax.swing.JLabel jLabel6;
private javax.swing.JLabel jLabel7;
private javax.swing.JRadioButton jRadioButton1;
private javax.swing.JRadioButton jRadioButton2;
private javax.swing.JRadioButton jRadioButton3;
private javax.swing.JRadioButton jRadioButton4;
private javax.swing.JRadioButton jRadioButton5;
// End of variables declaration
}

```

OUTPUT:



16.Design a JFrame to display name of various Solar eclipses phases in combo box, allow the user to choose a particular phase, on selection display appropriate images with its detailed information. (HW)

CODE:

```
package com.mycompany.swingprojects;

import javax.swing.*;
import java.awt.*;

public class solar_eclipse_phases extends javax.swing.JFrame {
    private static final java.util.logging.Logger logger = java.util.logging.Logger.getLogger(solar_eclipse_phases.class.getName());
    public solar_eclipse_phases() {
        initComponents();
    }
    @SuppressWarnings("unchecked")
    // <editor-fold defaultstate="collapsed" desc="Generated Code">
    private void initComponents() {
        jLabel1 = new javax.swing.JLabel();
        jScrollPane2 = new javax.swing.JScrollPane();
        jTextArea2 = new javax.swing.JTextArea();
        jComboBox1 = new javax.swing.JComboBox<>();
        jScrollPane1 = new javax.swing.JScrollPane();
        jTextArea1 = new javax.swing.JTextArea();
        jLabel2 = new javax.swing.JLabel();
        setDefaultCloseOperation(javax.swing.WindowConstants.EXIT_ON_CLOSE);
        jLabel1.setFont(new java.awt.Font("Segoe UI Variable", 1, 14)); // NOI18N
        jLabel1.setForeground(new java.awt.Color(51, 255, 51));
        jLabel1.setText("Solar Eclipses");
        jScrollPane2.setForeground(new java.awt.Color(51, 255, 51));
        jTextArea2.setBackground(new java.awt.Color(0, 0, 0));
        jTextArea2.setColumns(20);
        jTextArea2.setFont(new java.awt.Font("Segoe UI", 1, 14)); // NOI18N
        jTextArea2.setForeground(new java.awt.Color(102, 255, 102));
        jTextArea2.setLineWrap(true);
        jTextArea2.setRows(5);
        jTextArea2.setText("Solar eclipses occur when the Sun, the Moon, and Earth line up, either fully or partially. Depending on how they align, eclipses provide a unique, exciting view of either the Sun or the Moon.\n\nA solar eclipse happens when the Moon passes between the Sun and Earth, casting a shadow on Earth that either fully or partially blocks the Sun's light in some areas. This only happens occasionally, because the Moon doesn't orbit in the exact same plane as the Sun and Earth do. The time when they are aligned is known as eclipse season, which happens twice a year.");
        jTextArea2.setWrapStyleWord(true);
        jScrollPane2.setViewportView(jTextArea2);
        jComboBox1.setBackground(new java.awt.Color(0, 0, 0));
        jComboBox1.setFont(new java.awt.Font("Segoe UI Emoji", 1, 14)); // NOI18N
        jComboBox1.setForeground(new java.awt.Color(102, 255, 102));
        jComboBox1.setModel(new javax.swing.DefaultComboBoxModel<>(new String[] { "Choose Phases..", "1st Phase", "2nd Phase", "3rd Phase", "4th Phase" }));
        jComboBox1.addActionListener(new java.awt.event.ActionListener() {
            public void actionPerformed(java.awt.event.ActionEvent evt) {
                jComboBox1ActionPerformed(evt);
            }
        });
        jScrollPane1.setForeground(new java.awt.Color(51, 255, 51));
        jTextArea1.setBackground(new java.awt.Color(0, 0, 0));
        jTextArea1.setColumns(20);
        jTextArea1.setFont(new java.awt.Font("Segoe UI Emoji", 1, 14)); // NOI18N
        jTextArea1.setForeground(new java.awt.Color(51, 255, 51));
        jTextArea1.setLineWrap(true);
        jTextArea1.setRows(5);
        jTextArea1.setWrapStyleWord(true);
        jScrollPane1.setViewportView(jTextArea1);
        jLabel2.setBackground(new java.awt.Color(255, 255, 255));
        jLabel2.setForeground(new java.awt.Color(51, 255, 51));
        jLabel2.setPreferredSize(new java.awt.Dimension(300, 200));
        javax.swing.GroupLayout layout = new javax.swing.GroupLayout(getContentPane());
        getContentPane().setLayout(layout);
    }
}
```

```

layout.setHorizontalGroup(
    layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)
        .addGroup(layout.createSequentialGroup()
            .addGroup(layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)
                .addGap(26, 26, 26)
                .addGroup(layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)
                    .addGroup(layout.createSequentialGroup()
                        .addGroup(layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)
                            .addComponent(jComboBox1, javax.swing.GroupLayout.PREFERRED_SIZE,
                                javax.swing.GroupLayout.DEFAULT_SIZE, javax.swing.GroupLayout.PREFERRED_SIZE)
                            .addComponent(jImage, javax.swing.GroupLayout.PREFERRED_SIZE, 293,
                                javax.swing.GroupLayout.PREFERRED_SIZE))
                        .addGap(29, 29, 29)
                        .addComponent(jScrollPane1, javax.swing.GroupLayout.PREFERRED_SIZE, 283,
                                javax.swing.GroupLayout.PREFERRED_SIZE))
                            .addComponent(jScrollPane2, javax.swing.GroupLayout.PREFERRED_SIZE, 605,
                                javax.swing.GroupLayout.PREFERRED_SIZE)))
                    .addGroup(layout.createSequentialGroup()
                        .addGap(285, 285, 285)
                        .addComponent(jLabel1)))
                .addContainerGap(24, Short.MAX_VALUE))
        );
layout.setVerticalGroup(
    layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)
        .addGroup(layout.createSequentialGroup()
            .addContainerGap()
            .addComponent(jLabel1, javax.swing.GroupLayout.PREFERRED_SIZE, 57, javax.swing.GroupLayout.PREFERRED_SIZE)
            .addPreferredGap(javax.swing.LayoutStyle.ComponentPlacement.RELATED)
            .addComponent(jScrollPane2, javax.swing.GroupLayout.PREFERRED_SIZE, 128,
                javax.swing.GroupLayout.PREFERRED_SIZE)
            .addPreferredGap(javax.swing.LayoutStyle.ComponentPlacement.UNRELATED)
            .addGroup(layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING, false)
                .addGroup(layout.createSequentialGroup()
                    .addComponent(jComboBox1, javax.swing.GroupLayout.PREFERRED_SIZE, javax.swing.GroupLayout.DEFAULT_SIZE,
                        javax.swing.GroupLayout.PREFERRED_SIZE)
                    .addPreferredGap(javax.swing.LayoutStyle.ComponentPlacement.RELATED)
                    .addComponent(jImage, javax.swing.GroupLayout.PREFERRED_SIZE, 213, javax.swing.GroupLayout.PREFERRED_SIZE))
                .addComponent(jScrollPane1))
            .addContainerGap(24, Short.MAX_VALUE))
        );
jImage.getAccessibleContext().setAccessibleName("jImage");

pack();
}
private void jComboBox1ActionPerformed(java.awt.event.ActionEvent evt) {
    String s = (String) jComboBox1.getSelectedItem();
    switch (s) {
    case "1st Phase":
        jTextArea1.setText("The Moon fully blocks the Sun, darkening the sky like dawn or
dusk.\n"
        + "The Sun's corona becomes visible. Safe to briefly remove eclipse glasses only
during totality.");
        jImage.setIcon(new
javax.swing.ImageIcon("C:\\Users\\ROHIT\\OneDrive\\Desktop\\Swing Programs\\Total
Solar Eclipse.jpeg"));
        break;
    case "2nd Phase":
        jTextArea1.setText("The Moon is farther from Earth and appears smaller,\n"
        + "creating a bright ring ('ring of fire') around it.");

```

```
        jImage.setIcon(new
javax.swing.ImageIcon("C:\\Users\\ROHIT\\OneDrive\\Desktop\\Swing Programs\\Annual
Solar Eclipse.jpeg"));
        break;
    case "3rd Phase":
        jTextArea1.setText("The Sun, Moon, and Earth are not perfectly aligned,\n"
            + "so only part of the Sun is covered, forming a crescent shape.");
        jImage.setIcon(new
javax.swing.ImageIcon("C:\\Users\\ROHIT\\OneDrive\\Desktop\\Swing Programs\\Partial
Solar Eclipse.jpeg"));
        break;
    case "4th Phase":
        jTextArea1.setText("Shifts between total and annular along its path\n"
            + "due to Earth's curvature.");
        jImage.setIcon(new
javax.swing.ImageIcon("C:\\Users\\ROHIT\\OneDrive\\Desktop\\Swing Programs\\Hybrid
Solar Eclipse.jpeg"));
        break;
    default:
        jTextArea1.setText("");
        jImage.setIcon(null);
        break;
    }
}

public static void main(String args[]) {
    java.awt.EventQueue.invokeLater(() -> new solar_eclipse_phases().setVisible(true));
}

// Variables declaration - do not modify
private javax.swing.JComboBox<String> jComboBox1;
private javax.swing.JLabel jImage;
private javax.swing.JLabel jLabel1;
private javax.swing.JScrollPane jScrollPane1;
private javax.swing.JScrollPane jScrollPane2;
private javax.swing.JTextArea jTextArea1;
private javax.swing.JTextArea jTextArea2;
// End of variables declaration
}
```

OUTPUT:

Solar Eclipses

Solar eclipses occur when the Sun, the Moon, and Earth line up, either fully or partially. Depending on how they align, eclipses provide a unique, exciting view of either the Sun or the Moon.

A solar eclipse happens when the Moon passes between the Sun and Earth, casting a shadow on Earth that either fully or partially blocks the Sun's light in some areas. This

Choose Phases..

Choose Phases..

1st Phase

2nd Phase

3rd Phase

4th Phase

Solar Eclipses

Solar eclipses occur when the Sun, the Moon, and Earth line up, either fully or partially. Depending on how they align, eclipses provide a unique, exciting view of either the Sun or the Moon.

A solar eclipse happens when the Moon passes between the Sun and Earth, casting a shadow on Earth that either fully or partially blocks the Sun's light in some areas. This

1st Phase

The Moon fully blocks the Sun, darkening the sky like dawn or dusk. The Sun's corona becomes visible. Safe to briefly remove eclipse glasses only during totality.

Solar Eclipses

Solar eclipses occur when the Sun, the Moon, and Earth line up, either fully or partially. Depending on how they align, eclipses provide a unique, exciting view of either the Sun or the Moon.

A solar eclipse happens when the Moon passes between the Sun and Earth, casting a shadow on Earth that either fully or partially blocks the Sun's light in some areas. This

2nd Phase

The Moon is farther from Earth and appears smaller, creating a bright ring ('ring of fire') around it.

Solar Eclipses

Solar eclipses occur when the Sun, the Moon, and Earth line up, either fully or partially. Depending on how they align, eclipses provide a unique, exciting view of either the Sun or the Moon.

A solar eclipse happens when the Moon passes between the Sun and Earth, casting a shadow on Earth that either fully or partially blocks the Sun's light in some areas. This

3rd Phase

The Sun, Moon, and Earth are not perfectly aligned, so only part of the Sun is covered, forming a crescent shape.

Solar Eclipses

Solar eclipses occur when the Sun, the Moon, and Earth line up, either fully or partially. Depending on how they align, eclipses provide a unique, exciting view of either the Sun or the Moon.

A solar eclipse happens when the Moon passes between the Sun and Earth, casting a shadow on Earth that either fully or partially blocks the Sun's light in some areas. This

4th Phase

Shifts between total and annular along its path due to Earth's curvature.

17. Write a java program to display the content of the text file in the table using the following GUI.

CODE:

```
package com.mycompany.swingprojects;
```

```
import java.io.*;
```

```
import javax.swing.JFileChooser;
```

```
public class table_frame extends javax.swing.JFrame {
```

```
    private static final java.util.logging.Logger logger =
```

```
java.util.logging.Logger.getLogger(table_frame.class.getName());
```

```
public table_frame() {
    initComponents();
}
@SuppressWarnings("unchecked")
// <editor-fold defaultstate="collapsed" desc="Generated Code">
private void initComponents() {
    jScrollPane1 = new javax.swing.JScrollPane();
    jTable1 = new javax.swing.JTable();
    jButton1 = new javax.swing.JButton();
    setDefaultCloseOperation(javax.swing.WindowConstants.EXIT_ON_CLOSE);
    setTitle("Row and Column Converter");
    jTable1.setModel(new javax.swing.table.DefaultTableModel(
        new Object [][] {
            {null, null, null, null},
            {null, null, null, null},
            {null, null, null, null},
            {null, null, null, null}
        },
        new String [] {
            "Title 1", "Title 2", "Title 3", "Title 4"
        }
    ));
    jScrollPane1.setViewportView(jTable1);
    jButton1.setText("Open");
    jButton1.addActionListener(new java.awt.event.ActionListener() {
        public void actionPerformed(java.awt.event.ActionEvent evt) {
            jButton1ActionPerformed(evt);
        }
    });
    javax.swing.GroupLayout layout = new javax.swing.GroupLayout(getContentPane());
    getContentPane().setLayout(layout);
    layout.setHorizontalGroup(
        layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)
            .addGroup(layout.createSequentialGroup()
                .addContainerGap()
                .addComponent(jScrollPane1, javax.swing.GroupLayout.DEFAULT_SIZE, 594,
Short.MAX_VALUE)
                .addContainerGap())
            .addGroup(layout.createSequentialGroup()
                .addGap(262, 262, 262)
                .addComponent(jButton1)
                .addContainerGap(javax.swing.GroupLayout.DEFAULT_SIZE, Short.MAX_VALUE))
    );
    layout.setVerticalGroup(
        layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)
            .addGroup(layout.createSequentialGroup()
                .addContainerGap()
                .addComponent(jScrollPane1, javax.swing.GroupLayout.PREFERRED_SIZE, 192,
javax.swing.GroupLayout.PREFERRED_SIZE)
```

```
.addPreferredGap(javax.swing.LayoutStyle.ComponentPlacement.RELATED, 26, Short.MAX_VALUE)
.addComponent(jButton1)
.addGap(48, 48, 48))
);
pack();
}
private void jButton1ActionPerformed(java.awt.event.ActionEvent evt) {
    JFileChooser fc = new JFileChooser();
    int r = fc.showOpenDialog(this);
    if(r == JFileChooser.APPROVE_OPTION){
        int i;
        String p = fc.getSelectedFile().getPath();
        String msg = "";
        try{
            File f = new File(p);
            String head[] = {"Name", "Address", "Contact Number"};
            String data[][] = new String[4][3];
            String k;
            i=0;
            int j=0;
            FileReader fr = new FileReader(p);
            BufferedReader br = new BufferedReader(fr);
            while ((k = br.readLine()) != null) {
                if(i == 0){
                    data[j][0] = k;
                    i++;
                }else if(i == 1){
                    data[j][1] = k;
                    i++;
                }else if(i == 2){
                    data[j][2] = k;
                    i = 0;
                    j++;
                }
            }
            jTable1.setModel(new javax.swing.table.DefaultTableModel(data, head));
        }
        catch(Exception e){
            System.out.println(e);
        }
    }
}

public static void main(String args[]) {
    java.awt.EventQueue.invokeLater(() -> new table_frame().setVisible(true));
}
// Variables declaration - do not modify
private javax.swing.JButton jButton1;
private javax.swing.JScrollPane jScrollPane1;
private javax.swing.JTable jTable1;
// End of variables declaration
}
```

OUTPUT:

The screenshot displays two windows from a Java application. The left window, titled 'JTable.txt', shows a text file with the following content:

```
Ajay  
Mumbai  
1234567890  
Rita  
Goa  
9087654321  
Champak  
Andheri  
1243568790  
Doc  
Borivali  
1234677545
```

The right window, titled 'Row and Column Converter', displays a table with three columns: 'Name', 'Address', and 'Contact Number'. The table contains the following data:

Name	Address	Contact Number
Ajay	Mumbai	1234567890
Rita	Goa	9087654321
Champak	Andheri	1243568790
Doc	Borivali	1234677545

Below the table, there is an 'Open' button.